Designing Leadership under Social Conformity

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Abstract

This paper studies the optimal design of leading by example in a social network where players differ in individual productivity and incur a cost for deviating from the average effort of their neighbors (the social norm). Moving first exploits a norm feedback effect: by inducing followers to adjust, a leader reshapes her own social norm and eases peer pressure, which raises aggregate effort if and only if her productivity exceeds that norm. We construct an exact leading index to identify the single key leader and show that as conformity strengthens, selection shifts from local productivity advantages weighted by local reciprocity to a global comparison favoring the most productive player. With multiple leaders, an optimal group requires the remaining followers to form a dominating set; group selection reduces to an NP-hard weighted maximum directed cut problem under weak conformity, whereas under strong conformity a "less is more" principle collapses every optimal group to a single player with the highest productivity. Moreover, finer multi-stage hierarchies need not increase aggregate effort, and when leaders signal private information about an unknown state, retaining a productive player as a follower can outperform appointing her to lead.

Keywords: Leadership; Social conformity; Network games; Organizational design.

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1 Introduction

Mobilizing individual effort is central to collective performance, yet private incentives need not align with broader societal interests.[1] Leading by example offers an important mechanism for bridging this gap, even in the absence of contractual relationships or hierarchical authority (Hermalin, 1998; Güth et al., 2007). For instance, in Bolivian communities, asking elected leaders to donate publicly before other residents substantially increased total giving for schoolbooks (Jack and Recalde, 2015). During the American Civil War, military units whose officers led from the front and shared battlefield risks experienced lower desertion rates (Costa, 2014). A key channel behind the effectiveness of leadership is that the actions of early movers can establish a behavioral reference point, thereby shaping the incentives of those who follow (Acemoglu and Jackson, 2015; Tankard and Paluck, 2016). Although these observations underscore the economic significance of leadership, fundamental questions remain: how should leadership be organized, what determines the selection of key leaders, and to what extent does the surrounding social environment shape the outcomes of such leadership arrangements?

Social conformity—the tendency for individuals to align their behavior with that of their peers—is a prominent feature of many social interactions. In a variety of settings, prevailing practices and shared expectations can serve as social reference points that guide individual choices (Andreoni and Bernheim, 2009; Bursztyn et al., 2020). Existing research provides several behavioral foundations for this reliance, including social comparison (Festinger, 1954), social-image concerns (Bernheim, 1994; Bursztyn and Jensen, 2017), and identity-based preferences (Akerlof and Kranton, 2000).[2] In the context of leadership, the desire to conform has two contrasting implications. On the one hand, it makes followers more receptive to a leader's example; on the other hand, it discourages leaders from deviating from their peers. Effective leadership thus hinges on both the ability to influence others and the incentive to take the lead.

[1] This discrepancy arises in settings such as charitable fundraising (Frey and Meier, 2004; Shang and Croson, 2009), workplace production (Mas and Moretti, 2009; Bandiera et al., 2010), and school initiatives (Paluck and Shepherd, 2012; Paluck et al., 2016).

[2] Social comparison provides a basis for self-evaluation: individuals assess their opinions and abilities relative to those of others, particularly when objective standards are unavailable. Social-image concerns arise when individuals care about how others perceive them. Because observable actions convey information about underlying characteristics, individuals may conform to gain social approval or avoid disapproval. Identity-based preferences link individuals' utility to their social identities and associated behavioral prescriptions, making conformity to these prescriptions intrinsically rewarding and deviations psychologically costly.

Despite this tension, theory offers limited guidance on how to organize leadership while taking into account the dual role of conformity.

This paper studies the optimal design of leading by example in the presence of social conformity. Organizational members (hereafter players) interact on a social network and differ in individual productivity, which governs the intrinsic incentive to exert effort. Following Ushchev and Zenou (2020), we introduce a preference for conformity through local-average interactions: each player incurs a cost for deviating from the average effort of her neighbors, which serves as a *social norm*. In this setting, the game features both *payoff externalities*, as neighbors' efforts shift the social norm, and *strategic complementarities*, as higher peer effort weakly increases a player's marginal return to effort. We model leadership through the timing of effort choices. Our main analysis considers a two-stage sequential game in which an organizational planner selects a subset of players to move first as *leaders*, while the remaining players observe these choices and then respond as *followers*. The planner's objective is to maximize aggregate effort.

Our main findings are summarized as follows. First, we characterize the subgame-perfect Nash equilibrium (SPNE) for a given leader group. At any level of social conformity, each player's effort is a weighted average of individual productivities in the network, with weights jointly determined by the strength of conformity and the allocation of leadership. In contrast to the local-aggregate model of Zhou and Chen (2015),[3] we show that introducing leadership under social conformity can either increase or decrease individual and aggregate effort relative to the simultaneous-move Nash equilibrium.

The central mechanism stems from the dual role of social conformity. To see this, consider a single leader whose productivity exceeds the social norm, so that conformity initially depresses her Nash effort. Moving first allows the leader to exploit a *norm feedback* effect: anticipating that higher effort will induce followers to adjust upward, the leader can raise her own social norm and thereby ease the pressure to conform. These responses propagate through the network and increase aggregate effort. Conversely, when the leader's productivity falls below the social norm, the same mechanism works in reverse and reduces aggregate effort.

Second, we address the key-leader problem under social conformity: if the planner chooses a single player to move first, who should it be? We construct an exact leading index to identify

[3]Zhou and Chen (2015) establish that each player's equilibrium contribution is weakly higher under sequential moves than under simultaneous moves. The result is driven by positive network spillovers and strategic complementarities, whereby leaders anticipate that greater contributions induce followers to contribute more.

the key leader—the single leader who maximizes aggregate effort. Because effective leadership requires both a strong incentive to increase effort and the ability to propagate that increase through the network, the most productive player need not be the key leader. Moreover, the key leader in the presence of social norms can differ substantially from the key leader of Zhou and Chen (2015), even when simultaneous-move best responses coincide. This divergence arises because the marginal return to leading is proportional to Nash effort in the local-aggregate model, but to the difference between Nash effort and the social norm in the local-average model. Consequently, more productive peers raise a candidate's Nash effort but reduce her leadership gain by elevating the social norm she already faces.

We further show that the taste for conformity systematically reshapes the selection of the key leader. To understand this result, consider the extreme cases approaching *pure individualism* and *pure conformism*.[4] When the taste for conformity is sufficiently small, the effect of local connections dominates: the leadership gain is proportional to a player's productivity advantage over direct neighbors, weighted by *local reciprocity*—the average weight those neighbors assign to the leader. Because neighbors with fewer links dilute this influence less, a player who stands out locally can outperform a globally more productive peer. When the taste for conformity is sufficiently large, by contrast, indirect connections throughout the network come into play. Although Nash efforts approach a consensus given by the degree-weighted average of individual productivities, a single leader anchors the entire population at her own productivity. As a result, the most productive player must be the key leader. Between these two limits, leadership gains can vary nonmonotonically with the taste for conformity, and stronger conformity can reverse the ranking of candidates.

Third, we extend the analysis to multiple leaders and characterize optimal group selection. The key trade-off is that appointing more leaders provides more examples but leaves fewer followers to respond. Due to the combinatorial nature of group selection, simply ranking candidates by their individual leading indices is not sufficient to ensure a global optimum. Nevertheless, for any level of social conformity, we establish two structural properties of an optimal leader group. First, the group consists only of players whose productivity exceeds the social norm, or equivalently, who would increase aggregate effort if selected as a single leader.

[4]Under pure individualism, each player's equilibrium effort equals her individual productivity and is independent of the social norm, so leadership has no effect. However, under pure conformism, all players in the simultaneous-move benchmark choose the same effort level, equal to the degree-weighted average of individual productivities.

Second, the remaining followers must form a *dominating set* in the network, meaning that every leader has at least one follower neighbor. This result reveals a substitutability between connected leaders, who cannot respond to one another as followers and therefore make tightly clustered leadership less attractive.

Beyond these general characterizations, more explicit results obtain in the limiting cases of weak and strong conformity. For sufficiently small conformity, group selection among players whose productivity exceeds the average of their neighbors reduces to a weighted maximum directed cut problem. The planner seeks a partition that maximizes the total weight of links directed from leaders to followers, where each link is weighted by the leader's local productivity advantage and reciprocal influence. Because this program contains the classical maximum cut problem as a special case, optimal group selection under weak conformity is NP-hard in general. For sufficiently large conformity, by contrast, a "*less is more*" principle emerges: every optimal group collapses to a single player with the highest individual productivity. Whereas a single top leader pulls the entire network toward the maximum productivity, appointing additional leaders weakens norm feedback and reduces aggregate effort, even when all selected leaders are equally productive.

Moreover, we investigate the leadership design problem for specific network architectures. In the case of complete networks, we show that for any given group size, assigning players with higher productivity to the leader group is always beneficial. In addition, an optimal leader group shrinks monotonically as the taste for conformity strengthens. In complete bipartite networks, key-leader selection favors the part with higher maximum and average productivities, or the part with smaller size when these moments coincide. Furthermore, when one part is uniformly more productive, optimal group selection exhibits a bang-bang pattern: at a conformity threshold, the optimal leader group drops abruptly from that entire part to a singleton.

Fourth, we relax the two-stage restriction and allow the planner to organize leadership across an arbitrary sequence of stages. A natural question in organizational design is what constitutes the optimal hierarchy of leadership. Existing theories suggest that a more granular hierarchy—ultimately a sequential chain—always benefits the organization.[5] Instead, we show

[5]In the local-aggregate network model, Zhou and Chen (2015) show that, starting from any fixed sequence, aggregate effort always increases when a set of simultaneously moving players instead chooses effort sequentially, suggesting that any local modification toward the sequential-move game is beneficial. Similarly, in a team-production model with hierarchical signaling, Zhou (2016) shows that a full chain is the optimal hierarchy because it maximizes the incentive to lead by example across stages.

that in the presence of social conformity, finer sequencing need not increase aggregate effort. The main reason is that moving earlier allows a player to reshape the social norm through subsequent responses, raising effort only when her productivity exceeds that norm. Consequently, introducing intermediate tiers into the leadership hierarchy is valuable when a player should first follow a more productive neighbor and then lead a less productive one. To obtain more intuition, we consider the limiting cases of weak and strong conformity. If conformity is sufficiently weak, an optimal sequence assigns players with positive local productivity advantages to move one by one in decreasing order of those advantages. All remaining players then act simultaneously in the final stage. If conformity is sufficiently strong, by contrast, any multi-stage arrangement beyond a direct leader–follower hierarchy is suboptimal. As a result, a single player with the highest productivity moves first, and the rest of the players follow together.

Finally, we consider several extensions of our baseline model. In many organizational settings, leaders not only move first to set an example, but also hold superior information about an unknown state that affects the returns to effort (Hermalin, 1998; Zhou and Chen, 2016). We therefore allow leaders to privately observe the state before choosing effort, while followers infer the state from the leaders' actions. In a separating equilibrium, we show that signaling raises equilibrium effort above the complete-information benchmark, as revealing a favorable state induces followers to exert higher effort, which shifts the social norm upward and eases peer pressure for leaders whose effort exceeds that norm. At the same time, incomplete information introduces a novel trade-off in leadership design. Because more productive followers respond more strongly to the signaled state, appointing a productive player to lead sacrifices her strong response as a follower. Under weak conformity, the leadership gain is proportional to a leader's local productivity advantage, weighted by the *follower-to-leader productivity ratio*. Consequently, in a complete network the most productive player need not be selected to lead, and in a complete bipartite network an intermediate group size can become optimal; under strong conformity, by contrast, the informational gain vanishes and optimal leadership again collapses to a single player with the highest productivity.

We also examine how variations in the baseline specification of conformity preferences and network interactions affect our results. Specifically, we show that our main insights continue to hold when players have heterogeneous tastes for conformity and interact through weighted and directed networks. In addition, we study *anti-conformity*, whereby players benefit from

differentiating their efforts from the social norm. Although efforts become strategic substitutes, a unique equilibrium persists under a moderate level of anti-conformity and preserves our characterization of leader selection. Finally, we consider *shifted social norms* that set a player's reference level above or below the average effort of her neighbors. In this case, even under homogeneous productivities, a downward shift in the social norm enables leaders to raise aggregate effort, whereas an upward shift induces leaders to reduce aggregate effort.

1.1 Related Literature

Our study contributes to the literature on the optimal design of organizations and social networks. Specifically, we bridge (i) the organizational literature on leading by example and hierarchy design (Hermalin, 1998; Winter, 2010; Zhou and Chen, 2015; Zhou, 2016; Zhou and Chen, 2016) and (ii) the network game literature on peer effects and social norms (Ballester et al., 2006; Galeotti et al., 2010; Bramoullé et al., 2014; Ushchev and Zenou, 2020; Boucher et al., 2024).

In the organizational literature, existing theories show that early movers can motivate followers by generating positive network spillovers (Zhou and Chen, 2015), by expanding observability across hierarchical tiers (Winter, 2010; Zhou, 2016), or by signaling private information about common returns (Hermalin, 1998; Komai et al., 2007; Zhou and Chen, 2016; Zhou et al., 2023). Even when organizational conformity is present, as in Matsuno (2026), followers conform unilaterally to the leader's action. Throughout this work, influence is one-directional: leaders shape the incentives of followers without being constrained by how followers behave. Yet what happens when leaders are embedded in the same social network as followers and face peer pressure from the very norms they seek to move? Incorporating this mutual dependence overturns three canonical design principles. Leadership is no longer unconditionally beneficial; it raises aggregate effort only when a leader's productivity exceeds the social norm. Appointing more leaders or constructing a finer chain—the optimal architecture in Zhou and Chen (2015) and Zhou (2016)—can backfire by depriving leaders of responsive followers, culminating in a "*less is more*" principle under strong conformity. Moreover, when leaders hold private information, promoting the most productive member to lead can be suboptimal because doing so forfeits her strong response as a follower.

Our work also advances the literature on peer effects and social norms in networks. Seminal studies characterize simultaneous-move equilibria under local-aggregate spillovers (Ballester

et al., 2006; Galeotti et al., 2010; Bramoullé et al., 2014) and local-average social norms (Ushchev and Zenou, 2020; Boucher et al., 2024). In simultaneous play, these two microfoundations can generate identical linear best responses, differing only through a quadratic peer-norm term that each player treats as fixed. Does this observational equivalence carry over to sequential leadership design? We show that it does not. Once a leader moves first, followers' subsequent adjustments make the leader's social norm endogenous, activating the quadratic norm loss that is silent in simultaneous play. As a result, whereas the local-aggregate leading index rewards candidates with high Nash effort (Zhou and Chen, 2015), our local-average leading index rewards candidates whose Nash effort exceeds their social norm—so having more productive neighbors raises a player's Nash effort yet diminishes her value as a leader.

Finally, our analysis complements recent work on how prominent agents and policy interventions shape behavioral conventions. Prior studies examine how visible actions redirect expectations across generations (Acemoglu and Jackson, 2015), how imitation or preference adaptation governs norm dynamics (Goldbaum, 2019; Olcina et al., 2024), how stubborn leaders or norm recommendations sustain conventions (Cabrales and Hauk, 2024; Block et al., 2025), and how a planner can modify individual characteristics in simultaneous-move games (Meng, 2025). Instead of endowing leaders with fixed actions, norm-setting authority, or altered characteristics, we hold preferences and links fixed and use the timing of effort choices as the sole organizational instrument. A broader implication for organizational design is that leadership and followership are inseparable: because leaders rely on followers to shift the social norm, an optimal hierarchy must determine not only who takes the lead, but also who remains behind to respond.

The remainder of the paper is organized as follows. Section 2 presents the model, the simultaneous-move Nash benchmark, and the leadership design problem. Section 3 establishes the two-stage equilibrium characterization, analyzes key-leader and key-group selection, and derives explicit results for special network architectures. Section 4 concludes. All proofs are relegated to the Appendix.

2 Model

Our main departure from standard models of leading by example is that social conformity shapes the incentives of both leaders and followers. While followers adjust their efforts toward a

leader's example, leaders themselves face pressure to conform to a social norm that shifts with follower responses.

2.1 Preferences and the Environment

Consider an organization of $n \geq 2$ members (or players), indexed by $i \in N := \{1, \ldots, n\}$, who interact in a social network g. The adjacency matrix $\mathbf{G} = [g_{ij}]$ is an $(n \times n)$ matrix with $\{0, 1\}$ entries that keeps track of direct connections in the network: players i and j are directly connected if and only if $g_{ij} = 1$, and $g_{ij} = 0$ otherwise. We assume that g is connected, undirected ($g_{ij} = g_{ji}$), and has no self-loops ($g_{ii} = 0$). For each player i, define $N(i) := \{j \in N : g_{ij} = 1\}$ as the set of direct neighbors and $d_i := \sum_{j=1}^{n} g_{ij} > 0$ as player i's *degree*.

Let $\widehat{\mathbf{G}} = [\widehat{g}_{ij}]$ denote the $(n \times n)$ row-normalized adjacency matrix with entries $\widehat{g}_{ij} := g_{ij}/d_i$, where $\widehat{g}_{ij}$ is the weight that player i assigns to player j in forming a local reference group; every direct neighbor receives equal weight $1/d_i$, so each row of $\widehat{\mathbf{G}}$ sums to one.

Each player $i \in N$ is characterized by an exogenous *individual productivity* $\alpha_i > 0$ and chooses an effort level $x_i \geq 0$. Let $\boldsymbol{\alpha} := (\alpha_1, \ldots, \alpha_n)^\top$ and $\mathbf{x} := (x_1, \ldots, x_n)^\top$ denote the productivity and effort profiles. Player i's *social norm* is the average effort of her direct neighbors, $\bar{x}_i := \sum_{j=1}^{n} \widehat{g}_{ij} x_j$. Preferences take the linear-quadratic form of Ushchev and Zenou (2020):

$$u_i(x_i, \mathbf{x}_{-i}; g) := \alpha_i x_i - \frac{1}{2} x_i^2 - \frac{\gamma}{2}(x_i - \bar{x}_i)^2, \qquad \gamma \geq 0, \tag{1}$$

where $\gamma \geq 0$ measures the taste for conformity, and the network, productivities, and payoff parameters are common knowledge. In (1), $\alpha_i x_i - \frac{1}{2} x_i^2$ is the intrinsic (net) return from effort in the absence of peer effects. The *conformity cost* $-\frac{\gamma}{2}(x_i - \bar{x}_i)^2$ captures the cost of deviating from the social norm $\bar{x}_i$, which penalizes player i in proportion to the squared distance between own effort and the average effort of direct neighbors.[6] Throughout the analysis, we use $\lambda := \gamma/(1 + \gamma) \in [0, 1)$ to parameterize the relative strength of conformity.

Because the social norm $\bar{x}_i$ is formed from neighbors' efforts, each player's effort generates payoff externalities across links: for any $j \neq i$,

$$\frac{\partial u_i(x_i, \mathbf{x}_{-i}; g)}{\partial x_j} = \gamma \widehat{g}_{ij}(x_i - \bar{x}_i). \tag{2}$$

[6]The qualitative trade-offs in our analysis extend to a general convex conformity cost $\gamma\phi(x_i - \bar{x}_i)$, where $\phi(\cdot)$ is twice continuously differentiable and satisfies $\phi(0) = \phi'(0) = 0$ and $\phi''(z) > 0$ for all $z \in \mathbb{R}$.

When $\gamma > 0$ and $\widehat{g}_{ij} > 0$, higher effort by neighbor j raises player i's social norm $\bar{x}_i$. Whether this norm shift benefits or hurts player i depends on how x_i compares with $\bar{x}_i$: when $x_i > \bar{x}_i$, a higher $\bar{x}_i$ brings the social norm closer to player i's effort and reduces the conformity cost, whereas when $x_i < \bar{x}_i$, a higher $\bar{x}_i$ pushes the norm further above x_i and increases that cost. At the same time, efforts are strategic complements, since $\partial^2 u_i(x_i, \mathbf{x}_{-i}; g)/(\partial x_i \partial x_j) = \gamma \widehat{g}_{ij} \geq 0$ for any $j \neq i$: higher effort by a neighbor always raises player i's marginal return to effort. Together, these two features govern the incentives for leadership by example: when a leader commits to higher effort ahead of followers, strategic complementarity induces followers to raise their subsequent efforts, and the resulting upward shift in the leader's social norm rewards a leader whose effort exceeds the norm ($x_i > \bar{x}_i$) but penalizes a leader whose effort falls short of it ($x_i < \bar{x}_i$).[7]

We use the following notation throughout the paper.

- Vectors are columns, and $\top$ denotes transposition. For a vector $\mathbf{z}$ and a set $A \subseteq N$, $\mathbf{z}_A$ is the subvector indexed by A, and $\mathbf{z}_{-i}$ omits coordinate i. For a matrix $\mathbf{T} = [t_{ij}]$, $\mathbf{T}_{AB}$ retains the rows in A and columns in B, $\mathbf{T}_{-i}$ deletes row and column i, and for square $\mathbf{T}$, $\mathbf{T}^{\mathrm{D}}$ retains only the diagonal entries while $\rho(\mathbf{T})$ denotes the spectral radius.
- The vectors $\mathbf{1}_A$ and $\mathbf{0}_A$ have all entries equal to one and zero, respectively, and $\mathbf{I}_A$ is the identity matrix on coordinates in A (subscripts on $\mathbf{1}, \mathbf{0}, \mathbf{I}$ are omitted when clear from context). The vector $\mathbf{e}_i$ has entry one at coordinate i and zero elsewhere. For sets $A, B \subseteq N$, $|A|$ is the cardinality of A, and $A \setminus B$ is the set of elements in A outside B.
- Vector and matrix inequalities are componentwise: $\mathbf{A} \geq \mathbf{B}$ means that every entry of $\mathbf{A} - \mathbf{B}$ is nonnegative, and $\mathbf{A} > \mathbf{B}$ means that every entry is strictly positive.

2.2 The Organizational Problem

We model leading by example through the timing of effort choices. As in Zhou and Chen (2015), an organizational planner who knows the network and all individual productivities selects a

[7] In the standard local-aggregate model (see, for example, Ballester et al. (2006) and Zhou and Chen (2015)), payoffs take the form $u_i^{\mathrm{agg}}(x_i, \mathbf{x}_{-i}; g) = \alpha_i x_i - \frac{1}{2}x_i^2 + \delta x_i \sum_{j=1}^n g_{ij} x_j$, where $\delta \geq 0$ measures the strength of network interaction. Higher neighbor effort raises both the marginal return to own effort and the payoff level ($\partial u_i^{\mathrm{agg}}(x_i, \mathbf{x}_{-i}; g)/\partial x_j = \delta g_{ij} x_i \geq 0$) without creating a norm-deviation penalty. In Section 3, we compare leadership in the two models.

group of early movers before effort decisions take place in two stages; Figure 1 summarizes the timeline of the game.

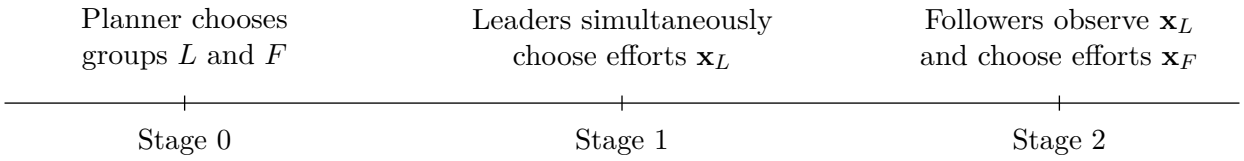


Figure 1: Timeline of the organizational problem.

In Stage 0, the planner publicly partitions the players into a leader group $L \subseteq N$ and a follower group $F := N \setminus L$. When players are ordered so that leaders precede followers, the adjacency matrix $\mathbf{G}$ and the row-normalized matrix $\widehat{\mathbf{G}}$ admit the block decompositions

$$\mathbf{G} = \begin{pmatrix} \mathbf{G}_{LL} & \mathbf{G}_{LF} \\ \mathbf{G}_{FL} & \mathbf{G}_{FF} \end{pmatrix}, \qquad \text{and} \qquad \widehat{\mathbf{G}} = \begin{pmatrix} \widehat{\mathbf{G}}_{LL} & \widehat{\mathbf{G}}_{LF} \\ \widehat{\mathbf{G}}_{FL} & \widehat{\mathbf{G}}_{FF} \end{pmatrix}. \tag{3}$$

In Stage 1, leaders in L simultaneously commit to public, irreversible efforts $\mathbf{x}_L$, taking into account the induced Stage 2 response of followers. In Stage 2, followers in F observe $\mathbf{x}_L$ and simultaneously choose their efforts $\mathbf{x}_F$. Payoffs are realized according to (1) at the final effort profile.

For a leader group $L \subseteq N$, let $\mathbf{x}^S(L)$ denote the subgame-perfect equilibrium effort profile in the two-stage game and $X^S(L) := \sum_{i=1}^n x_i^S(L)$ denote the corresponding aggregate effort. The planner selects a leader group to maximize aggregate equilibrium effort:

$$\max_{L \subseteq N} X^S(L) = \sum_{i=1}^n x_i^S(L). \tag{4}$$

We refer to a leader group L^* that solves (4) as an *optimal leader group* (or a *key group*). To measure the effect of leadership relative to the simultaneous-move benchmark, let $\mathbf{x}^N$ denote the simultaneous-move Nash equilibrium (hereafter the *Nash equilibrium*) effort profile and $X^N := \sum_{i=1}^n x_i^N$ the Nash-benchmark aggregate effort. The *leadership gain* of group L is defined as $\ell(L) := X^S(L) - X^N$. Because X^N does not depend on L, solving the organizational problem in (4) is equivalent to maximizing the leadership gain $\ell(L)$.

2.3 Simultaneous-move Benchmark

In the simultaneous-move benchmark (which arises when $L = \varnothing$ or $L = N$), all players move at the same time. Each player $i \in N$ chooses $x_i \geq 0$ to maximize (1), taking the efforts $\mathbf{x}_{-i}$ of all other players—and hence the social norm $\bar{x}_i$—as given. Because (1) is strictly concave in x_i, the first-order condition $(\alpha_i - x_i) - \gamma(x_i - \bar{x}_i) = 0$, or equivalently $x_i + \gamma(x_i - \bar{x}_i) = \alpha_i$, characterizes player i's unique best response. With $\lambda = \gamma/(1+\gamma)$, this condition yields the linear best-response function

$$x_i = (1-\lambda)\alpha_i + \lambda \bar{x}_i. \tag{5}$$

Each player's best response is a convex combination of own productivity α_i and the social norm $\bar{x}_i$, with weights $1-\lambda$ and λ that balance intrinsic effort incentives against the desire to conform. A stronger taste for conformity shifts weight toward the social norm and increases responsiveness to neighbors' efforts; because $\alpha_i > 0$, the best response is strictly positive for every nonnegative peer-effort profile and every $\lambda \in [0, 1)$. In the limiting case $\lambda = 0$ (*pure individualism*), each player ignores peer behavior and chooses $x_i = \alpha_i$; as $\lambda \to 1$ (*pure conformism*), each player seeks solely to match the social norm.

In vector form, the social norms are $\bar{\mathbf{x}} := \widehat{\mathbf{G}}\mathbf{x}$, so the best-response system (5) writes as $\mathbf{x} = (1-\lambda)\boldsymbol{\alpha} + \lambda\widehat{\mathbf{G}}\mathbf{x}$. Proposition 1 of Ushchev and Zenou (2020) characterizes the resulting Nash equilibrium.

Lemma 1 (Simultaneous-move equilibrium)**.** *The simultaneous-move game admits a unique Nash equilibrium, which is given by*

$$\mathbf{x}^N = \mathbf{M}^N \boldsymbol{\alpha}, \tag{6}$$

where $\mathbf{M}^N = [m_{ij}^N]$ *is an* $(n \times n)$ *simultaneous-move influence matrix defined as*[8]

$$\mathbf{M}^N = (1-\lambda)(\mathbf{I} - \lambda\widehat{\mathbf{G}})^{-1} = (1-\lambda)\sum_{t=0}^{\infty} \lambda^t \widehat{\mathbf{G}}^t. \tag{7}$$

Proof: See Appendix A.1.

[8]The adjacency matrix $\widehat{\mathbf{G}}$ is row-normalized, so $\rho(\widehat{\mathbf{G}}) = 1$. For $0 \leq \lambda < 1$, $\rho(\lambda\widehat{\mathbf{G}}) = \lambda < 1$, ensuring that the inverse exists and admits the Neumann series in (7).

By (6), player i's Nash effort can be written as $x_i^N = \sum_{j=1}^n m_{ij}^N \alpha_j$, where $m_{ij}^N = \partial x_i^N / \partial \alpha_j$ measures the marginal effect of player j's productivity on player i's equilibrium effort. As in Ushchev and Zenou (2020), the Neumann series in (7) expands this cross-effect across walk lengths:

$$m_{ij}^N = \frac{\partial x_i^N}{\partial \alpha_j} = \sum_{t=0}^{\infty} \underbrace{(1-\lambda)\lambda^t}_{\text{geometric weight}} \, (\widehat{\mathbf{G}}^t)_{ij},$$

where $(\widehat{\mathbf{G}}^t)_{ij}$ records the normalized weight of length-t walks from player i to player j, and the coefficients $(1-\lambda)\lambda^t$ form a geometric distribution over walk lengths $t \geq 0$.

Because $\widehat{\mathbf{G}}$ is row-stochastic, the influence matrix $\mathbf{M}^N$ is also row-stochastic, and connectedness ensures $m_{ij}^N > 0$ for all $i, j \in N$ whenever $0 < \lambda < 1$. Therefore, each player's Nash effort is a strictly positive convex combination of all individual productivities. When productivities are homogeneous ($\alpha_i = \alpha$ for all $i \in N$), row-stochasticity implies $x_i^N = \alpha$ for every player regardless of network position; unlike the local-aggregate model, network structure generates effort dispersion in Nash equilibrium only when productivities are heterogeneous.

In the absence of conformity ($\gamma = 0$), each player chooses her standalone effort, so $x_i^N = \alpha_i$ for all $i \in N$. The presence of conformity ($\gamma > 0$), however, creates a gap between player i's productivity α_i and Nash effort x_i^N, proportional to her deviation from the equilibrium social norm $\bar{x}_i^N := \sum_{j=1}^n \widehat{g}_{ij} x_j^N$:

$$\alpha_i - x_i^N = \gamma(x_i^N - \bar{x}_i^N) = \lambda(\alpha_i - \bar{x}_i^N). \tag{8}$$

Under heterogeneous productivities, the three differences in (8) share the same sign and can be either positive or negative depending on how player i's productivity compares with the social norm. When $\alpha_i > \bar{x}_i^N$, peer pressure constrains player i and depresses her effort below α_i ($\bar{x}_i^N < x_i^N < \alpha_i$); when $\alpha_i < \bar{x}_i^N$, peer pressure instead disciplines player i and raises her effort above α_i ($\alpha_i < x_i^N < \bar{x}_i^N$). In particular, because $\min_{j \in N} \alpha_j < \bar{x}_i^N < \max_{j \in N} \alpha_j$ for all $i \in N$ when $\gamma > 0$, a player with the highest productivity always exceeds the social norm, whereas a player with the lowest productivity always falls below it.

3 Leadership and Conformity

3.1 Equilibrium Characterization

For a leader group $L \subseteq N$, we solve for equilibrium efforts by backward induction, starting with follower responses to leading efforts. The block expressions in this subsection apply to $0 < |L| < n$; the empty and full groups reproduce the Nash equilibrium.

Followers. In Stage 2, each follower $k \in F$ observes the leading effort vector $\mathbf{x}_L$ and chooses $x_k \geq 0$ to maximize (1), taking the efforts of all other players as given. By (5), follower k's best response is $x_k = (1-\lambda)\alpha_k + \lambda \bar{x}_k$, where the social norm is decomposed into the committed efforts of leader neighbors and the efforts of follower neighbors:

$$\bar{x}_k = \sum_{i \in L} \widehat{g}_{ki} x_i + \sum_{j \in F} \widehat{g}_{kj} x_j.$$

In vector form, under the block partition in (3), the follower best responses satisfy $\mathbf{x}_F = (1-\lambda)\boldsymbol{\alpha}_F + \lambda \widehat{\mathbf{G}}_{FL}\mathbf{x}_L + \lambda \widehat{\mathbf{G}}_{FF}\mathbf{x}_F$. Define $\mathbf{U}_F := (\mathbf{I}_F - \lambda \widehat{\mathbf{G}}_{FF})^{-1}$. For a nonnegative effort vector $\mathbf{x}_L$, the follower subgame admits a unique equilibrium with strictly positive efforts,

$$\mathbf{x}_F(\mathbf{x}_L) = \mathbf{U}_F\Big[(1-\lambda)\boldsymbol{\alpha}_F + \lambda \widehat{\mathbf{G}}_{FL}\mathbf{x}_L\Big]. \tag{9}$$

Leaders. In Stage 1, each leader $k \in L$ anticipates how $\mathbf{x}_L$ shapes follower responses $\mathbf{x}_F(\mathbf{x}_L) = \Big(x_j(\mathbf{x}_L)\Big)_{j \in F}$ in (9) and hence the social norm against which leader k's own effort is evaluated:

$$\bar{x}_k(\mathbf{x}_L) = \sum_{i \in L} \widehat{g}_{ki} x_i + \sum_{j \in F} \widehat{g}_{kj} x_j(\mathbf{x}_L).$$

Because $g_{kk} = 0$ and $\mathbf{x}_F(\mathbf{x}_L)$ is linear in $\mathbf{x}_L$, leader k's commitment x_k affects $\bar{x}_k(\mathbf{x}_L)$ entirely through follower responses at the constant rate $c_k := \partial \bar{x}_k(\mathbf{x}_L)/\partial x_k = \sum_{j \in F} \widehat{g}_{kj}\, \partial x_j(\mathbf{x}_L)/\partial x_k$. A unit increase in x_k therefore raises leader k's norm by c_k and increases the deviation $x_k - \bar{x}_k(\mathbf{x}_L)$ by $1 - c_k$. Let $\mathbf{x}_{-k}(\mathbf{x}_L) := (\mathbf{x}_{L\setminus\{k\}}, \mathbf{x}_F(\mathbf{x}_L))$. With follower responses given by (9), leader k's

payoff in (1) is strictly concave in x_k, and the first-order condition decomposes the marginal payoff into three terms:

$$\frac{du_k\left(x_k, \mathbf{x}_{-k}(\mathbf{x}_L); g\right)}{dx_k} = \underbrace{\alpha_k - x_k}_{\substack{\text{standalone} \\ \text{productivity effect}}} - \underbrace{\gamma(x_k - \bar{x}_k)}_{\substack{\text{direct} \\ \text{social-norm effect}}} + \underbrace{\gamma c_k(x_k - \bar{x}_k)}_{\substack{\text{indirect} \\ \text{norm-feedback effect}}} = 0. \tag{10}$$

When leader k increases effort x_k, her marginal payoff is affected in three ways. First, there is a *standalone productivity effect*, $\alpha_k - x_k$, which is the marginal net return to effort in the absence of peer interactions. Second, there is a *direct social-norm effect*, $-\gamma(x_k - \bar{x}_k)$, which captures the marginal cost of departing from the social norm when $\bar{x}_k$ is held fixed. Whereas a follower in Stage 2 takes $\bar{x}_k$ as given and faces only these first two effects, a leader in Stage 1 also internalizes a third channel: the *indirect norm-feedback effect*, $\gamma c_k(x_k - \bar{x}_k)$, which arises because follower responses shift $\bar{x}_k$ in the direction of x_k at rate $c_k \geq 0$. For a leader, the indirect norm-feedback effect always acts in the *opposite* direction to the direct social-norm effect, reflecting the dual role of social conformity.

Let $\bar{x}_k^S$ and $\bar{x}_k^N$ denote player k's equilibrium social norm in the two-stage and simultaneous-move games (we omit the argument L when the leader group is fixed). By (10), each leader $k \in L$ chooses an equilibrium effort that is a weighted average of individual productivity and the final social norm:

$$x_k^S = \frac{1-\lambda}{1-\lambda c_k}\,\alpha_k + \frac{\lambda(1-c_k)}{1-\lambda c_k}\,\bar{x}_k^S. \tag{11}$$

Under social conformity, compared with the best response in the Nash equilibrium (5), moving first allows the leader to shift weight from the social norm toward own productivity. Individual productivity exceeds (falls below) the social norm if and only if the leader's effort lies below (above) individual productivity, that is, $\bar{x}_k^S < x_k^S < \alpha_k$ ($\alpha_k < x_k^S < \bar{x}_k^S$).[9]

In vector form, the leaders' social norms are $\bar{\mathbf{x}}_L(\mathbf{x}_L) := \left(\bar{x}_k(\mathbf{x}_L)\right)_{k\in L} = \widehat{\mathbf{G}}_{LL}\mathbf{x}_L + \widehat{\mathbf{G}}_{LF}\mathbf{x}_F(\mathbf{x}_L)$. By (9), we obtain

$$\bar{\mathbf{x}}_L(\mathbf{x}_L) = (1-\lambda)\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\boldsymbol{\alpha}_F + \mathbf{C}_L\mathbf{x}_L, \tag{12}$$

[9] Analogously to (8), rearranging (11) yields $\alpha_k - x_k^S = \gamma(1-c_k)(x_k^S - \bar{x}_k^S) = \frac{\lambda(1-c_k)}{1-\lambda c_k}(\alpha_k - \bar{x}_k^S)$. Because $0 \leq c_k < 1$, the three differences share the same sign whenever $\gamma > 0$.

with the *norm feedback matrix*

$$\mathbf{C}_L := (c_{ij})_{i,j\in L} = \frac{\partial \bar{\mathbf{x}}_L(\mathbf{x}_L)}{\partial \mathbf{x}_L} = \widehat{\mathbf{G}}_{LL} + \lambda \widehat{\mathbf{G}}_{LF}\mathbf{U}_F\widehat{\mathbf{G}}_{FL}, \tag{13}$$

where $\widehat{\mathbf{G}}_{LL}$ represents the direct influence of other leaders' commitments on the social norm and $\lambda\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\widehat{\mathbf{G}}_{FL}$ represents the indirect feedback through follower responses. Because the network has no self-loops, each diagonal entry $c_k := c_{kk}$ arises entirely through followers:

$$c_k = \frac{\partial \bar{x}_k(\mathbf{x}_L)}{\partial x_k} = \lambda(\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\widehat{\mathbf{G}}_{FL})_{kk} = \sum_{t=0}^{\infty} \lambda^{t+1}\Big(\widehat{\mathbf{G}}_{LF}\widehat{\mathbf{G}}_{FF}^{\,t}\widehat{\mathbf{G}}_{FL}\Big)_{kk}. \tag{14}$$

Each term records return walks that leave leader k for a follower neighbor, take t steps within the follower subnetwork, and return to k. The response matrices $\lambda\mathbf{U}_F\widehat{\mathbf{G}}_{FL}$ and $\mathbf{C}_L$ are nonnegative, with $\mathbf{C}_L\mathbf{1}_L \le \mathbf{1}_L$ and $0 \le c_k \le \lambda < 1$. Rewriting (11) in vector form gives

$$(1-\lambda)\mathbf{x}_L^S + \lambda(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})(\mathbf{x}_L^S - \bar{\mathbf{x}}_L^S) = (1-\lambda)\boldsymbol{\alpha}_L.$$

Together with $\bar{\mathbf{x}}_L^S = \bar{\mathbf{x}}_L(\mathbf{x}_L^S)$ from (12), define

$$\mathbf{V}_L := \Big[(1-\lambda)\mathbf{I}_L + \lambda(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})(\mathbf{I}_L - \mathbf{C}_L)\Big]^{-1}. \tag{15}$$

Then the equilibrium leading efforts are

$$\mathbf{x}_L^S = (1-\lambda)\mathbf{V}_L\Big[\boldsymbol{\alpha}_L + \lambda(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\boldsymbol{\alpha}_F\Big].$$

Theorem 1 (Subgame-perfect equilibrium)**.** *For any leader group $L \subseteq N$, the two-stage game admits a unique subgame-perfect Nash equilibrium. With leaders ordered first, equilibrium efforts are given by*

$$\begin{pmatrix}\mathbf{x}_L^S\\ \mathbf{x}_F^S\end{pmatrix} = \mathbf{M}^S(L)\begin{pmatrix}\boldsymbol{\alpha}_L\\ \boldsymbol{\alpha}_F\end{pmatrix}, \tag{16}$$

where $\mathbf{M}^S(L) = [m_{ij}^S(L)]$ is an $(n \times n)$ two-stage influence matrix defined as

$$\mathbf{M}^S(L) := (1-\lambda)\begin{pmatrix}\mathbf{V}_L & \lambda\mathbf{V}_L(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\\ \lambda\mathbf{U}_F\widehat{\mathbf{G}}_{FL}\mathbf{V}_L & \mathbf{U}_F + \lambda^2\mathbf{U}_F\widehat{\mathbf{G}}_{FL}\mathbf{V}_L(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\end{pmatrix}. \tag{17}$$

Proof: See Appendix A.2.

The entry $m_{ij}^S(L) = \partial x_i^S(L)/\partial \alpha_j$ measures the marginal effect of player j's productivity on player i's equilibrium effort (with the argument L omitted when the leader group is fixed). For $0 < \lambda < 1$, all entries of $\mathbf{M}^S(L)$ are strictly positive, so a higher productivity for any player raises effort throughout the network. As in the simultaneous-move model of Ushchev and Zenou (2020), $\mathbf{M}^S(L)$ is row-stochastic: each equilibrium effort $x_i^S(L) = \sum_{j=1}^n m_{ij}^S(L)\alpha_j$ is a convex combination of productivities bounded between $\min_{j\in N} \alpha_j$ and $\max_{j\in N} \alpha_j$. If players are homogeneous ($\alpha_i = \alpha$ for all $i \in N$), then $x_i^S(L) = x_i^N = \alpha$ for all $i \in N$ and $L \subseteq N$, so $\ell(L) = 0$: leadership alters effort only when productivities are heterogeneous.

When productivities are heterogeneous, the block structure of $\mathbf{M}^S(L)$ in (17) reflects how leadership interacts with the dual role of social conformity. On the one hand, follower conformity transmits leaders' commitments across the follower subnetwork through $\lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL}$ and generates norm feedback $\mathbf{C}_L$, which attenuates leaders' effective pressure to conform by $(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})$ in $\mathbf{V}_L$ and shifts equilibrium weight toward leaders' own productivities $\boldsymbol{\alpha}_L$. On the other hand, leaders continue to face peer pressure from both other leaders and followers, so follower productivities $\boldsymbol{\alpha}_F$ shape leading commitments through $(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})\widehat{\mathbf{G}}_{LF}\mathbf{U}_F$ and spill back over to the follower group. Unlike the local-aggregate model of Zhou and Chen (2015), where first-mover commitments weakly raise all efforts under sufficiently weak interactions, row-stochasticity of $\mathbf{M}^S(L)$ and $\mathbf{M}^N$ implies that leadership under social conformity redistributes productivity weights and can either increase or decrease equilibrium efforts relative to the Nash benchmark.

Proposition 1 (Equilibrium effort comparison)**.** *For a leader group $L \subseteq N$, leadership increases the equilibrium effort of player $j \in N$ relative to the Nash benchmark ($x_j^S > x_j^N$) if and only if*

$$\sum_{i\in L} m_{ji}^S c_i(\alpha_i - x_i^N) > 0. \tag{18}$$

Proof: See Appendix A.3.

To see why $c_i(\alpha_i - x_i^N)$ governs leader i's impact in (18), evaluate leader i's Stage 1 marginal payoff in (10) at the Nash equilibrium $\mathbf{x}^N$, where the standalone productivity effect and the direct social-norm effect cancel by (8), leaving only the indirect norm-feedback effect:

$$\left.\frac{du_i\left(x_i, \mathbf{x}_{-i}(\mathbf{x}_L); g\right)}{dx_i}\right|_{\mathbf{x}=\mathbf{x}^N} = \underbrace{(\alpha_i - x_i^N) - \gamma(x_i^N - \bar{x}_i^N)}_{=0} + \gamma c_i(x_i^N - \bar{x}_i^N) = c_i(\alpha_i - x_i^N).$$

The difference $\alpha_i - x_i^N$ between productivity and Nash effort therefore determines the direction of leader i's first-mover incentive, norm feedback c_i scales its magnitude, and the equilibrium weight m_{ji}^S transmits the resulting adjustment to player j (with the reverse strict inequality in (18) characterizing $x_j^S < x_j^N$). With multiple leaders, opposing incentives can offset one another and generate different net effects across players. However, in the case of a single leader, only one term enters (18) and all equilibrium efforts move in the same direction.

Remark 1. For a single leader $L = \{i\}$, equilibrium efforts move in the same direction across the network, increasing relative to the Nash benchmark if and only if $\alpha_i > x_i^N > \bar{x}_i^N$, or equivalently,

$$\alpha_i > \sum_{j \neq i} \frac{m_{ij}^N}{1 - m_{ii}^N} \alpha_j. \tag{19}$$

Proof: See Appendix A.4.

Since $\sum_{j \neq i} m_{ij}^N / (1 - m_{ii}^N) = 1$, the right-hand side of (19) is a weighted average of all other players' productivities, with weights determined by their simultaneous-move influence on player i. A single leader whose productivity exceeds this average productivity is constrained by a lower social norm in the Nash equilibrium ($\alpha_i > x_i^N > \bar{x}_i^N$), so moving first increases her effort and induces higher efforts from all followers; conversely, a leader whose productivity falls below this average productivity is disciplined by a higher social norm in the Nash equilibrium ($\alpha_i < x_i^N < \bar{x}_i^N$), so moving first reduces her effort and lowers all others' efforts.

3.2 Single Leader

We begin with the case where the organizational planner selects a single leader $L = \{i\}$. The planner then solves the *key leader problem*, defined as $\max_{i \in N} X^S(\{i\})$, and a player who solves this problem is referred to as a *key leader*. For notational simplicity, we write $\ell_i := \ell(\{i\}) = X^S(\{i\}) - X^N$ for the leadership gain of a single leader i.

The choice of a single leader trades off the candidate's incentive to adjust effort against the network propagation of that adjustment. When $L = \{i\}$, let $x_j(x_i)$ denote follower j's response in (9). Each follower $j \neq i$ responds at the constant rate $\partial x_j(x_i)/\partial x_i = m_{ji}^N / m_{ii}^N$, and leader i's norm feedback is

$$c_i = \frac{m_{ii}^N + \lambda - 1}{\lambda m_{ii}^N} \in (0, \lambda]. \tag{20}$$

Together with $x_i^N = \sum_{k=1}^n m_{ik}^N \alpha_k$, (20) implies that the simultaneous-move influence matrix $\mathbf{M}^N = [m_{ij}^N]$ pins down follower responsiveness, leader i's norm feedback c_i (which increases in m_{ii}^N), and the gap between productivity and Nash effort.

Proposition 2 (Key leader under conformity)**.** *A key leader i^* maximizes the leadership gain*

$$\ell_i = \frac{(1-\lambda)c_i}{1-\lambda c_i(2-c_i)} \frac{\sum_{j=1}^n m_{ji}^N}{m_{ii}^N} (\alpha_i - x_i^N). \tag{21}$$

We refer to ℓ_i as the local-average leading index *of player i.*

Proof: See Appendix A.5.

The leading index ℓ_i decomposes into the leader's own effort adjustment $x_i^S - x_i^N$ and a network propagation multiplier. In the first factor, $(1-\lambda)c_i$ captures the marginal incentive to adjust effort through norm feedback, while the denominator $1-\lambda c_i(2-c_i) = (1-\lambda)+\lambda(1-c_i)^2$ reflects the curvature of the leader's objective, combining intrinsic effort-cost curvature $(1-\lambda)$ with the residual norm-deviation curvature $\lambda(1-c_i)^2$. The second factor, $\sum_{j=1}^n m_{ji}^N/m_{ii}^N = 1+\sum_{j\neq i} \partial x_j(x_i)/\partial x_i$, aggregates the resulting follower responses across the network. Because both network factors are strictly positive, $\text{sign}(\ell_i) = \text{sign}(\alpha_i - x_i^N)$: under heterogeneous productivities, a player with the highest productivity always raises aggregate effort ($\ell_i > 0$), whereas a player with the lowest productivity lowers it ($\ell_i < 0$). Among candidates with $\alpha_i > x_i^N$, however, a more productive player can be a weaker leader than a candidate whose network position generates stronger return feedback and broader follower propagation.

The role of social conformity. How a candidate's leading index $\ell_i(\lambda)$ varies with the taste for conformity depends on the peer benchmark embedded in $x_i^N(\lambda)$. As conformity strengthens, that benchmark shifts from the local neighbor average $\bar{\alpha}_i := \sum_{j=1}^n \hat{g}_{ij}\alpha_j$ to the degree-weighted network average $\bar{\alpha}_g := \sum_{i=1}^n \pi_i \alpha_i$, where $\pi_i := d_i / \sum_{j=1}^n d_j$.

Corollary 1 (Key leader under weak and strong conformity)**.**

(i) *For sufficiently small conformity, a key leader maximizes $r_i(\alpha_i - \bar{\alpha}_i)$, where $r_i := (\widehat{\mathbf{G}}^2)_{ii} = \frac{1}{d_i}\sum_{j\in N(i)} \frac{1}{d_j}$.*

(ii) *For sufficiently large conformity, a key leader is a player with the highest productivity.*

Proof: See Appendix A.6.

In part (i), when the taste for conformity is sufficiently small ($\lambda \to 0$), the gap between productivity and Nash effort satisfies $\alpha_i - x_i^N = \lambda(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^2)$. Meanwhile, expanding (14) for $L = \{i\}$ gives

$$c_i = \lambda(\widehat{\mathbf{G}}^2)_{ii} + \sum_{t=1}^{\infty} \lambda^{t+1} \left(\widehat{\mathbf{G}}_{iF} \widehat{\mathbf{G}}_{FF}^t \widehat{\mathbf{G}}_{Fi} \right) = \lambda \sum_{j \in N(i)} \widehat{g}_{ij} \widehat{g}_{ji} + \mathcal{O}(\lambda^2) = \lambda r_i + \mathcal{O}(\lambda^2).$$

Substituting these expansions into (21) yields $\ell_i(\lambda) = \lambda^2 r_i(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^3)$. Define $r_{ij} := \widehat{g}_{ij}\widehat{g}_{ji} = g_{ij}/(d_i d_j)$, so that $r_i = \sum_{j \in N(i)} r_{ij}$; we refer to r_{ij} as the *link reciprocity* between players i and j, and to r_i as the *local reciprocity* of player i.[10] For leader i's neighbor $j \in N(i)$, $\widehat{g}_{ji} = 1/d_j$ represents the weight that neighbor j places on player i's effort, so $r_i = d_i^{-1} \sum_{j \in N(i)} \widehat{g}_{ji}$ equals the average weight that i's neighbors assign to i in their own social norms. In a largely individualist organization, the planner therefore favors a candidate with a large local productivity advantage $\alpha_i - \bar{\alpha}_i$ and high local reciprocity r_i: when a candidate's neighbors have fewer links, those neighbors do not dilute i's influence across many peers and thus return stronger feedback.

In part (ii), when the taste for conformity is sufficiently large ($\lambda \to 1$), Proposition 3 of Ushchev and Zenou (2020) shows that Nash efforts converge to the degree-weighted average productivity, $\lim_{\lambda \to 1} x_j^N(\lambda) = \bar{\alpha}_g$ for all $j \in N$.[11] When player i moves first, however, followers match the social norm one-for-one as $\lambda \to 1$ ($c_i \to 1$ and $m_{ji}^N/m_{ii}^N \to 1$), which eliminates the leader's residual pressure to conform and anchors all efforts at α_i, that is, $\lim_{\lambda \to 1} x_j^S(\{i\}) = \alpha_i$ for all $j \in N$. The limiting leadership gain is therefore $\lim_{\lambda \to 1} \ell_i(\lambda) = n(\alpha_i - \bar{\alpha}_g)$: because the Nash benchmark $\bar{\alpha}_g$ is common to all players, the ranking of candidates depends only on individual productivity α_i, and a player with the highest productivity is selected as the key leader.

Between these two limits, Corollary 2 characterizes how the leading index responds to individual productivities, the taste for conformity, and follower links.

Corollary 2 (Comparative statics)**.** *The local-average leading index ℓ_i of a player $i \in N$ satisfies the following properties:*

[10] In the probabilistic interpretation of the local-average model (Ushchev and Zenou (2020)), where $\widehat{\mathbf{G}} = [\widehat{g}_{ij}]$ is the transition probability matrix of a Markov chain on N, $r_{ij} = \widehat{g}_{ij}\widehat{g}_{ji}$ is the probability of moving from state i to state j and returning to i in the next step, and $r_i = \sum_{j \in N(i)} r_{ij} = (\widehat{\mathbf{G}}^2)_{ii}$ is the two-step return probability of state i.

[11] Because g is connected, all degree weights $\pi_i = d_i / \sum_{j=1}^n d_j$ lie in $(0, 1)$ and sum to one. Under heterogeneous productivities, $\bar{\alpha}_g$ is therefore a strict convex combination satisfying $\min_{i \in N} \alpha_i < \bar{\alpha}_g < \max_{i \in N} \alpha_i$.

(i) Productivities. *It strictly increases in player i's own productivity α_i but strictly decreases in any follower's productivity α_j $(j \neq i)$.*

(ii) Conformity. *If $\bar{\alpha}_i < \alpha_i \leq \bar{\alpha}_g$, ℓ_i is nonmonotone in λ and attains an interior global maximum; if $\bar{\alpha}_g \leq \alpha_i < \bar{\alpha}_i$, it attains an interior global minimum.*

(iii) Link dilution. *If $\alpha_i > \bar{\alpha}_i$, adding a new link between a neighbor $j \in N(i)$ and another follower strictly decreases ℓ_i for all sufficiently small λ.*

Proof: See Appendix A.7.

In part (i), a higher own productivity α_i widens the gap between productivity and Nash effort because x_i^N responds less than one for one ($m_{ii}^N < 1$); a higher follower productivity α_j $(j \neq i)$ instead raises x_i^N without changing α_i and compresses the additional effort induced by commitment (even though aggregate effort $X^S(\{i\})$ increases in α_j because $m_{kj}^S(\{i\}) > 0$). In part (ii), when α_i lies between the weak-conformity benchmark $\bar{\alpha}_i$ and the strong-conformity benchmark $\bar{\alpha}_g$, the difference $\alpha_i - x_i^N(\lambda)$ changes sign as λ increases from 0 to 1, which generates an interior extremum for $\ell_i(\lambda)$. In part (iii), whereas adding links in local-aggregate games always increases walk counts and raises leadership gains, an additional link between a neighbor $j \in N(i)$ and another follower raises d_j, dilutes the norm weight $1/d_j$ that neighbor j places on leader i, and lowers local reciprocity r_i without altering $\alpha_i - \bar{\alpha}_i$.

Comparison with the local-aggregate model. Expanding the quadratic conformity cost in (1) separates player i's payoff into a local-aggregate component and a norm-level term:

$$u_i(x_i, \mathbf{x}_{-i}; g) = \alpha_i x_i - \frac{1}{2}x_i^2 - \frac{\gamma}{2}(x_i - \bar{x}_i)^2 = \underbrace{\alpha_i x_i - \frac{1+\gamma}{2}x_i^2 + \gamma x_i \sum_{j=1}^{n} \widehat{g}_{ij} x_j}_{=: u_i^{\text{agg}}(x_i, \mathbf{x}_{-i}; g)} - \frac{\gamma}{2}\bar{x}_i^2, \tag{22}$$

where $u_i^{\text{agg}}(x_i, \mathbf{x}_{-i}; g)$ is a local-aggregate payoff.[12] Because the norm-level term $-\gamma\bar{x}_i^2/2$ does not depend on own effort x_i, the local-average and local-aggregate games generate the same simultaneous-move best responses, Nash equilibrium $\mathbf{x}^N$, and Stage 2 follower responses (Ushchev and Zenou (2020)). Under leadership, however, a first mover anticipates that follower responses will shift the social norm $\bar{x}_i$, so the norm-level term enters the Stage 1 commitment problem.

[12] Dividing $u_i^{\text{agg}}(x_i, \mathbf{x}_{-i}; g)$ by $1+\gamma$ gives the standard local-aggregate payoff $(1-\lambda)\alpha_i x_i - \frac{1}{2}x_i^2 + \lambda x_i \sum_{j=1}^{n} \widehat{g}_{ij} x_j$ with row-normalized adjacency matrix $\widehat{\mathbf{G}}$ (Ushchev and Zenou (2020)). Throughout our comparisons, the local-aggregate model (or benchmark) refers to u_i^{agg} in (22).

Under the local-aggregate payoff $u_i^{\text{agg}}(x_i, \mathbf{x}_{-i}; g)$ in (22), the aggregate effort gain from appointing player i as a single leader is given by the *local-aggregate leading index* (Zhou and Chen (2015)):

$$\mathcal{L}_i = \frac{\lambda c_i}{1 - 2\lambda c_i} \frac{\sum_{j=1}^n m_{ji}^N}{m_{ii}^N} x_i^N, \tag{23}$$

which is well defined when $1 - 2\lambda c_i > 0$. To compare the two models directly, using (8) rewrites the local-average leading index in (21) in terms of the difference between Nash effort and the social norm:

$$\ell_i = \frac{\lambda c_i}{1 - 2\lambda c_i + \lambda c_i^2} \frac{\sum_{j=1}^n m_{ji}^N}{m_{ii}^N} (x_i^N - \bar{x}_i^N). \tag{24}$$

Comparing (23) and (24) highlights two economic differences. First, leadership in the local-aggregate model is driven by the candidate's Nash effort level, whereas leadership under conformity is driven by the gap between Nash effort and the social norm. In the local-aggregate game, more productive neighbors raise x_i^N and strengthen candidate i's incentive to lead; under social conformity, more productive neighbors also elevate the social norm $\bar{x}_i^N$, which narrows that gap and weakens the gain from appointing candidate i. Second, the denominator of ℓ_i contains the additional curvature term λc_i^2: because follower responses shift the leader's own norm, the norm-level cost tempers the scale of commitment and ensures a finite equilibrium effort for all $\lambda \in (0, 1)$.

Example 1. To see how conformity can reverse single-leader selection relative to the local-aggregate benchmark, consider the bridge network illustrated in Figure 2, where two triangles $\{1, 2, 3\}$ and $\{4, 5, 6\}$ are joined by the bridge $\{3, 4\}$ and the productivity profile is $\boldsymbol{\alpha} = (\alpha_B, \alpha_A, \alpha_A, \alpha_A, \alpha_A, \alpha_B)^\top$ with $\alpha_A > \alpha_B > 0$. Let $A = \{2, 3, 4, 5\}$ and $B = \{1, 6\}$ denote the high- and low-productivity groups, respectively. The low-productivity peripheral players $1, 6$ satisfy $x_i^N > \alpha_B$ and have $\ell_i < 0$ for all $\lambda \in (0, 1)$, so the planner chooses between the high-productivity *non-bridge* players $2, 5$ and the equally productive *bridge* players $3, 4$.

Solving the simultaneous-move benchmark yields the Nash efforts and their differences from the social norm for candidates 2 and 3:

$$x_2^N = \alpha_A - \frac{6\lambda}{(\lambda+2)(\lambda+6)}(\alpha_A - \alpha_B), \qquad x_2^N - \bar{x}_2^N = \frac{6(1-\lambda)}{(\lambda+2)(\lambda+6)}(\alpha_A - \alpha_B),$$
$$x_3^N = \alpha_A - \frac{2\lambda}{\lambda+6}(\alpha_A - \alpha_B), \qquad x_3^N - \bar{x}_3^N = \frac{2(1-\lambda)}{\lambda+6}(\alpha_A - \alpha_B).$$

Because the bridge link connects player 3 to another high-productivity player ($\alpha_4 = \alpha_A$), player 3 has a strictly higher Nash effort ($x_3^N > x_2^N$) for all $\lambda \in (0,1)$, as well as stronger norm feedback ($c_3 > c_2$) and network propagation ($\sum_{j=1}^6 m_{j3}^N/m_{33}^N > \sum_{j=1}^6 m_{j2}^N/m_{22}^N$). In (23), all three factors of $\mathcal{L}_i$ therefore favor *bridge* players throughout the admissible range of λ. Under social conformity, however, the high-productivity neighbor elevates player 3's social norm $\bar{x}_3^N$ even more than x_3^N, so *non-bridge* player 2 exceeds the social norm by a strictly larger margin, with $(x_2^N - \bar{x}_2^N)/(x_3^N - \bar{x}_3^N) = 3/(\lambda+2) > 1$ for all $\lambda \in (0,1)$.

Without loss of generality, take $\alpha_A = 2$ and $\alpha_B = 1$.[13] Let us first assume that the taste for conformity is relatively weak, say $\lambda = 0.2$. Evaluating (23) and (24) gives

$$\ell_2(0.2) \approx 0.0079 > 0.0068 \approx \ell_3(0.2), \qquad \mathcal{L}_2(0.2) \approx 0.0428 < 0.0508 \approx \mathcal{L}_3(0.2).$$

When conformity is weak, player 2's larger gap between effort and the social norm outweighs player 3's network advantage, so the local-average game selects a *non-bridge* player while the local-aggregate benchmark selects a *bridge* player.

Assume now that the taste for conformity is stronger, say $\lambda = 0.5$. As λ increases, this relative gap $3/(\lambda+2)$ shrinks from 3/2 toward 1, and the bridge player's stronger norm feedback and network propagation dominate in both models:

$$\ell_2(0.5) \approx 0.0510 < 0.0574 \approx \ell_3(0.5), \qquad \mathcal{L}_2(0.5) \approx 0.5252 < 0.7272 \approx \mathcal{L}_3(0.5).$$

[13] For any $\alpha_A > \alpha_B > 0$ and fixed $\lambda \in (0,1)$, ℓ_i in (24) scales linearly with $\alpha_A - \alpha_B$, so the ranking between $\ell_2(\lambda)$ and $\ell_3(\lambda)$ depends only on λ; in (23), $x_3^N > x_2^N$ holds for all $\alpha_A > \alpha_B > 0$, so $\mathcal{L}_3(\lambda) > \mathcal{L}_2(\lambda)$ throughout the admissible range of λ.

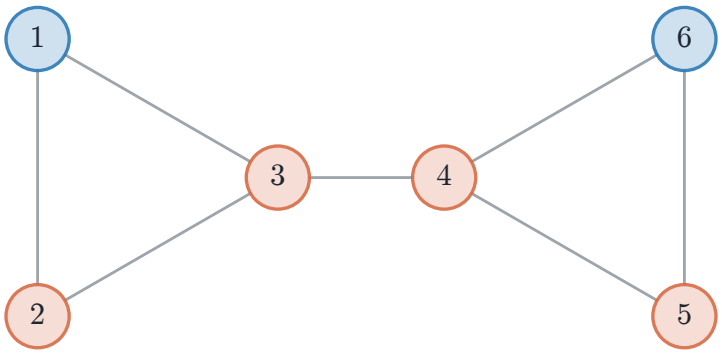


Figure 2: A bridge network

Note: Red and blue nodes indicate high and low individual productivities, respectively.

3.3 Multiple Leaders

We now allow the planner to select multiple players into the leader group to solve (4), which we refer to as the *key group problem*; we call a group L^* that solves (4) an *optimal leader group* (or a *key group*). Under social conformity, group selection entails two trade-offs that do not arise in the single-leader problem. First, whereas all players have a positive first-mover incentive in the local-aggregate model, under conformity a leader with productivity below the social norm adjusts effort downward and can offset the positive commitments of other leaders. Second, because leaders move simultaneously in Stage 1, promoting an additional player into L adds a first-mover commitment but removes a follower from F, which weakens the norm feedback c_i to neighboring leaders and leaves those leaders more constrained by the social norm. Selecting a key group therefore requires screening out low-productivity candidates while balancing the breadth of early commitments against the preservation of follower responses.

Definition 1 (Dominating set)**.** A set $S \subseteq N$ is a *dominating set* if each player outside S has at least one neighbor in S, that is, $N(i) \cap S \neq \varnothing$ for $i \in N \setminus S$.

Proposition 3 (Leader eligibility and follower domination)**.** *Suppose that productivities are heterogeneous. For any optimal leader–follower partition* (L^*, F^*) *with* $F^* := N \setminus L^*$*:*

(i) *The leader group* L^* *consists only of players with a positive local-average leading index* $(\ell_i > 0)$*.*

(ii) *The follower group* F^* *forms a* dominating set*.*

Proof: See Appendix A.8.

In part (i), by Remark 1 and Proposition 2, $\ell_i > 0$ is equivalent to $\alpha_i > x_i^N > \bar{x}_i^N$. To understand this result, consider a leader in L whose productivity lies at or below the social norm, i.e., $\alpha_i \leq \bar{x}_i^S(L)$, so that $\alpha_i \leq x_i^S(L)$. Moving first allows such a player to exploit norm feedback in the downward direction and reduce her effort. Reassigning any such leaders to the follower group eliminates their downward adjustments, strengthens norm feedback to the remaining leaders, and raises effort across the network. At an optimum, every leader therefore remains below her own productivity while raising efforts above the Nash benchmark, which yields $\bar{x}_i^N < x_i^N < x_i^S(L^*) < \alpha_i$ for all $i \in L^*$. In particular, a direct implication of part (i) is that assigning a least productive player to the leader group is never optimal.

Whereas part (i) restricts the leader group to candidates with $\ell_i > 0$, part (ii) implies that not all such candidates can lead together: even a high-productivity player may need to remain in the follower group so that F^* forms a dominating set. To see why, suppose that a leader has all of her neighbors in the leader group and none in the follower group. Because all of her neighbors act simultaneously with her, none of them can adjust to her effort subsequently; moving first therefore gives her no norm feedback and leaves her effort rule identical to that of a follower. Reassigning that player to the follower group does not alter her own response rule, but allows her to respond to her neighboring leaders, which strengthens their norm feedback and raises effort across the network. Follower domination consequently precludes appointing a leader together with all of her neighbors. Because both parts of Proposition 3 rely only on reassigning a leader to the follower group, the same conditions continue to hold under any upper bound on the leader group size.

Figure 3 provides an example to illustrate this tension. In the 3-tier tree shown in the figure, individual productivities decrease from the top player 1 (α_1) through the middle-tier players 2 and 3 ($\alpha_2 = \alpha_3$) to the bottom players 4 and 5 ($\alpha_4 = \alpha_5$), and it can be readily shown that the leading indices satisfy $\ell_1 > \ell_2 = \ell_3 > 0 > \ell_4 = \ell_5$. Consider first the key leader problem in panel (a): because player 1 has the highest leading index, player 1 is selected as the unique key leader. Consider next the key group problem in panel (b). Although players 1, 2, and 3 all have positive leading indices, appointing all three would leave player 1 connected only to co-leaders, violating the dominating structure of part (ii). Instead, the optimal leader group consists of the

middle-tier players $L^* = \{2, 3\}$, while the most productive player 1 joins the bottom players in the follower group $F^* = \{1, 4, 5\}$ to form a dominating set.

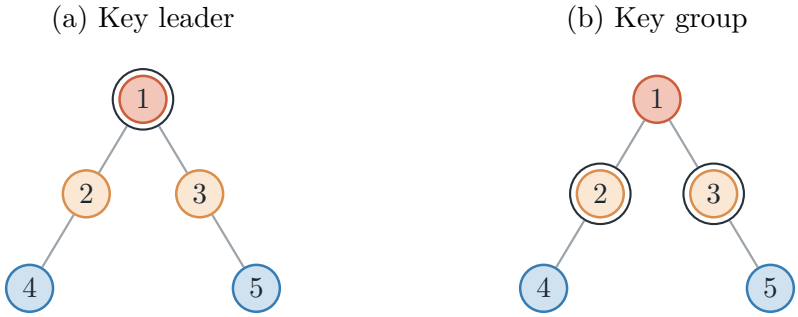


Figure 3: Key leader and key group in a 3-tier tree

Note: Red, orange, and blue nodes indicate high, medium, and low individual productivities, respectively; an outer ring identifies a leader. Panel (a) shows the unique key leader $L^* = \{1\}$, and panel (b) shows the unique key group $L^* = \{2, 3\}$.

Proposition 4 (Key groups under weak and strong conformity)**.** *Suppose that productivities are heterogeneous.*

(i) *For sufficiently small conformity, an optimal leader group maximizes*

$$\Lambda(L) := \sum_{i \in L} \sum_{j \in F} r_{ij}(\alpha_i - \bar{\alpha}_i). \tag{25}$$

(ii) *For sufficiently large conformity, an optimal leader group is a singleton consisting of a player with the highest productivity.*

Proof: See Appendix A.9.

When the taste for conformity is sufficiently small ($\lambda \to 0$), it is readily shown that the leadership gain of a group $L \subseteq N$ satisfies $\ell(L) = \lambda^2 \sum_{i \in L} \sum_{j \in F} \widehat{g}_{ij} \widehat{g}_{ji} (\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^3) = \lambda^2 \Lambda(L) + \mathcal{O}(\lambda^3)$. The value of leadership therefore arises at second order exclusively from links directed from leaders to followers, where each cross-group link $(i, j) \in L \times F$ contributes $\widehat{g}_{ij} \widehat{g}_{ji} (\alpha_i - \bar{\alpha}_i) = r_{ij}(\alpha_i - \bar{\alpha}_i)$ and links within L or within F contribute nothing.

Let $N_+ := \{i \in N : \alpha_i > \bar{\alpha}_i\}$ denote the set of candidates with positive local productivity gaps. Selecting an optimal leader group when the taste for conformity is sufficiently small reduces to a *weighted maximum directed cut* (MAX-DiCUT) problem among candidates in N_+, with noncandidates fixed as followers: the planner seeks a partition that maximizes the total

weight $w_{ij} := r_{ij}(\alpha_i - \bar{\alpha}_i)$ of links directed from $L \subseteq N_+$ to $F = N \setminus L$. Unlike the undirected maximum-cut problem in the local-aggregate model of Zhou and Chen (2015), each link here is valued by the local productivity advantage at its leading endpoint, so reversing the leader and follower roles of two adjacent players alters the gain. Nevertheless, when candidates in N_+ share the same local productivity advantage and link reciprocities coincide, maximizing $\Lambda(L)$ contains the classical maximum cut problem as a special case.

Claim 1. *Solving the key group problem in (25) is NP-hard in general. On the candidate domain 2^{N_+}, the second-order objective $\Lambda(L)$ is submodular:*[14] *the marginal gain from appointing a candidate falls strictly whenever an adjacent candidate also leads.*

Proof: See Appendix A.10.

To see the source of submodularity, rewrite $\Lambda(L)$ for any candidate group $L \subseteq N_+$ using local reciprocity r_i:

$$\Lambda(L) = \sum_{i \in L} r_i(\alpha_i - \bar{\alpha}_i) - \sum_{\substack{i,j \in L \\ i<j}} r_{ij}[(\alpha_i - \bar{\alpha}_i) + (\alpha_j - \bar{\alpha}_j)]. \tag{26}$$

The first sum adds the candidates' standalone second-order gains $\Lambda(\{i\}) = r_i(\alpha_i - \bar{\alpha}_i)$, while the second sum deducts the two follower responses lost whenever adjacent candidates move together in Stage 1 instead of responding to each other. Because this pairwise loss can outweigh a candidate's standalone gain, an optimal leader group may keep a high-index candidate as a follower rather than ranking leaders by individual indices ℓ_i alone. Moreover, since submodularity makes each candidate's marginal contribution lowest when all other candidates lead, appointing the entire candidate set N_+ maximizes Λ if and only if, for each $i \in N_+$,

$$(\alpha_i - \bar{\alpha}_i) \sum_{j \notin N_+} r_{ij} \geq \sum_{j \in N_+} r_{ij}(\alpha_j - \bar{\alpha}_j). \tag{27}$$

Condition (27) requires candidate i's leading contribution through noncandidate followers (left-hand side) to exceed the feedback that adjacent candidate leaders forfeit when i leads rather than follows (right-hand side). In particular, if N_+ is an *independent set* (so candidates share no links), the right-hand side vanishes and all candidates in N_+ are appointed; conversely, if

[14] A set function $\eta : 2^V \to \mathbb{R}$ is *submodular* if $\eta(S) + \eta(T) \geq \eta(S \cup T) + \eta(S \cap T)$ for all $S, T \subseteq V$, or equivalently if the marginal contribution $\eta(S \cup \{i\}) - \eta(S)$ decreases with S.

some candidate $i \in N_+$ has only candidate neighbors, that is, $N(i) \subseteq N_+$, the left-hand side vanishes and a key group must leave at least one member of N_+ as a follower.

When the taste for conformity is sufficiently large ($\lambda \to 1$), let $H := \arg\max_{i \in N} \alpha_i$ denote the set of most productive players. Whereas a single leader in H brings the entire network toward the maximum productivity $\max_{i \in N} \alpha_i$, shared leadership retains mutual norm constraints among leaders. With $c_i^\infty := \lim_{\lambda \to 1} c_i \in [0, 1)$, for any leader group L with $2 \leq |L| < n$, equilibrium efforts converge across the network to the common limit

$$x^\infty(L) := \frac{\displaystyle\sum_{i \in L} \frac{\pi_i}{1 - c_i^\infty} \alpha_i + \sum_{j \in F} \pi_j \alpha_j}{\displaystyle\sum_{i \in L} \frac{\pi_i}{1 - c_i^\infty} + \sum_{j \in F} \pi_j}. \tag{28}$$

Because all players receive strictly positive weight in (28), $x^\infty(L)$ lies strictly below $\max_{i \in N} \alpha_i$. A single leader in H therefore achieves a strictly higher limiting effort, and because the number of groups is finite, this advantage persists for λ sufficiently close to one (with finite-conformity gains breaking ties within H). Consequently, when $|H| \geq 2$ under heterogeneous productivities, appointing even a second leader from H lowers effort throughout the network: for sufficiently strong conformity, any $i \in H$ and any group $L \subseteq H$ with $|L| \geq 2$ satisfy $x_j^S(\{i\}) > x_j^S(L)$ for all $j \in N$.

Example 2. Consider the 3-regular network in Figure 4, where the inner triangle $A = \{1, 2, 3\}$ and the outer triangle $B = \{4, 5, 6\}$ are joined by matching links $\{1, 4\}, \{2, 5\}, \{3, 6\}$. Players in A have productivity α_A, and players in B have productivity α_B, with $\alpha_A > \alpha_B > 0$. Because all players have degree three ($r_{ij} = 1/9$ on each link and $r_i = 1/3$), players $i \in A$ have $\alpha_i - \bar{\alpha}_i = (\alpha_A - \alpha_B)/3 > 0$ (and $\alpha_i > x_i^N$), whereas players $j \in B$ have $\alpha_j - \bar{\alpha}_j = -(\alpha_A - \alpha_B)/3 < 0$ (and $\alpha_j < x_j^N$). By Proposition 3(i), an optimal leader group under social conformity consists only of players in the high-productivity triangle ($L^* \subseteq A$).

First, when the taste for conformity is sufficiently small ($\lambda \to 0$), for a group $L \subseteq A$ of size $l := |L| \in \{1, 2, 3\}$, the pairwise-loss decomposition (26) gives

$$\lim_{\lambda \to 0} \frac{\ell(L)}{\lambda^2} = l \, \frac{\alpha_A - \alpha_B}{9} - \binom{l}{2} \frac{2(\alpha_A - \alpha_B)}{27} = \frac{\alpha_A - \alpha_B}{27} \, l(4 - l).$$

This second-order gain is maximized at $l = 2$ (such as $L = \{1, 2\}$); appointing the third high-productivity player 3 eliminates follower responses on $\{1, 3\}$ and $\{2, 3\}$ and lowers aggregate effort, with $\lim_{\lambda\to 0}[\ell(\{1, 2, 3\}) - \ell(\{1, 2\})]/\lambda^2 = -(\alpha_A - \alpha_B)/27 < 0$.

Second, let $\mathcal{L}(L)$ denote the leadership gain in the local-aggregate game (22). Adding the nonadjacent low-productivity player $6 \in B$ to $\{1, 2\}$ creates three new leader–follower links $(\{6, 3\}, \{6, 4\}, \{6, 5\})$ without severing any link from $\{1, 2\}$ to followers:

$$\lim_{\lambda\to 0} \frac{\ell(\{1, 2, 6\}) - \ell(\{1, 2\})}{\lambda^2} = -\frac{\alpha_A - \alpha_B}{9} < 0, \qquad \lim_{\lambda\to 0} \frac{\mathcal{L}(\{1, 2, 6\}) - \mathcal{L}(\{1, 2\})}{\lambda^2} = \frac{\alpha_B}{3} > 0.$$

Because $\alpha_B > 0$, the additional cut links make $\{1, 2, 6\}$ optimal in the local-aggregate benchmark for small λ, whereas under conformity player 6 faces a higher social norm ($\alpha_6 < \bar{\alpha}_6$) and reduces aggregate effort when appointed.

Third, when the taste for conformity is sufficiently large ($\lambda \to 1$), a single leader $\{1\}$ anchors all efforts at α_A, whereas under $\{1, 2\}$ the two leaders have limiting return feedback $c_1^\infty = c_2^\infty = 1/3$ and (28) yields the common effort limit $(4\alpha_A + 3\alpha_B)/7$. Since $X^N = 3(\alpha_A + \alpha_B)$, the limiting leadership gains satisfy

$$\lim_{\lambda\to 1} \ell(\{1, 2\}) = \frac{3}{7}(\alpha_A - \alpha_B) < 3(\alpha_A - \alpha_B) = \lim_{\lambda\to 1} \ell(\{1\}).$$

Set $\alpha_A = 2$ and $\alpha_B = 1$, as in Figure 4: as λ increases from 0.01 to 0.75, the optimal leader group under conformity contracts from $\{1, 2\}$ to $\{1\}$, whereas the key group in the local-aggregate benchmark shifts from $\{1, 2, 6\}$ to $\{1, 6\}$ and retains a low-productivity co-leader.

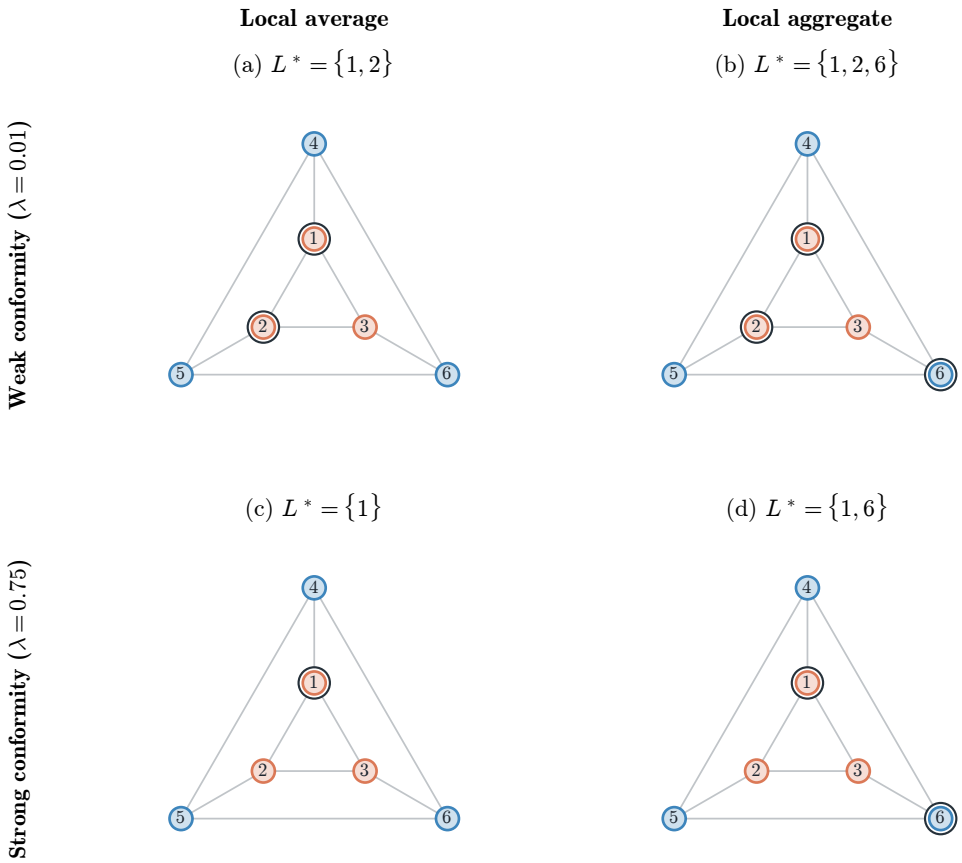


Figure 4: Key groups in a 3-regular network

Note: Red and blue nodes indicate high and low individual productivities, respectively; an outer ring identifies a leader. Columns compare key groups in the local-average game and the local-aggregate benchmark under weak conformity ($\lambda = 0.01$, panels (a) and (b)) and strong conformity ($\lambda = 0.75$, panels (c) and (d)).

3.4 Special Networks

Although solving the key group problem is NP-hard in general networks, regular interaction structures make group selection tractable and isolate the roles of productivity and group size. We examine two canonical architectures: complete networks and complete bipartite networks.

3.4.1 Complete networks

In small project teams, partnerships, or flat committees, members interact directly with one another and observe the average effort of the rest of the team. We model such an environment as the complete network K_n, where $g_{ij} = 1$ for all $i \neq j$. Even though network positions are symmetric, heterogeneous productivities $\alpha_1 > \cdots > \alpha_n > 0$ differentiate members' incentives to

lead. Define $\bar{\alpha} := n^{-1}\sum_{i=1}^{n}\alpha_i$ as the average productivity and $L_m := \{1,\ldots,m\}$ as the group of the m most productive players, and write $\bar{m} := \lfloor n/2 \rfloor$ and $\eta := (n-1+\lambda)/\lambda$.

Proposition 5 (Leader selection in complete networks)**.** *In the complete network K_n with $\alpha_1 > \cdots > \alpha_n > 0$:*

(i) Key leader. *Player 1 is the unique key leader.*

(ii) Key group. *An optimal leader group consists of the m^* most productive players, L_{m^*}, where $1 \leq m^* \leq \bar{m}$ is given by solving*

$$\max_{m\in\{1,\ldots,\bar{m}\}} \ell(L_m) = \frac{n(n-m)}{(n-1)\eta(\eta-n+m)-(n-m)^2}\left(\sum_{i=1}^{m}\alpha_i - m\bar{\alpha}\right). \tag{29}$$

(iii) Contraction with conformity. *For $0 < \lambda < \lambda' < 1$, optimal leader groups satisfy $L^*(\lambda') \subseteq L^*(\lambda)$.*

Proof: See Appendix A.12.

In part (i), because all members occupy identical positions in K_n, candidates share the same capacity to propagate effort to followers and receive the same norm feedback c_i, so single-leader selection reduces to choosing the member whose productivity exceeds the social norm (and hence Nash effort) by the largest margin. In part (ii), any fixed group size m selects the m most productive members L_m, and the objective in (29) captures how choosing m balances the cumulative productivity advantage $\sum_{i=1}^{m}(\alpha_i - \bar{\alpha})$ of early movers against the size $n-m$ of the follower group. Appointing an additional leader adds another high-productivity commitment, but also converts a responsive follower into a simultaneous first mover and weakens the norm feedback to existing leaders; an above-average member ($\alpha_i > \bar{\alpha}$) may therefore contribute more by following more productive colleagues than by moving in Stage 1, and the optimal leader group never exceeds half the team ($m^* \leq \bar{m}$).

Comparing consecutive group sizes in (29) shows that marginal additions to the leader group satisfy a single-crossing property: once adding the next most productive player fails to raise aggregate effort, any further expansion lowers aggregate effort.

Claim 2. *For $1 \leq m < \bar{m}$, including player $m+1$ in the leader group L_m increases aggregate effort, $\ell(L_{m+1}) > \ell(L_m)$, if and only if*

$$\alpha_{m+1} > \bar{\alpha} + \frac{\ell(L_m)}{n}\left[1 + \frac{(n-1)\eta^2}{(n-m)(n-m-1)}\right], \tag{30}$$

where $\ell(L_m)$ is given by (29). Moreover, if $\ell(L_{m+1}) \leq \ell(L_m)$, then $\ell(L_{s+1}) < \ell(L_s)$ for all $m < s < \bar{m}$.

Proof: See Appendix A.13.

Condition (30) requires candidate $m+1$'s productivity advantage over the team mean $\bar{\alpha}$ to exceed the per-capita leadership gain $\ell(L_m)/n$ scaled by the marginal loss of follower feedback. Because the right-hand side of (30) increases with m while α_{m+1} decreases, the planner can find the key group in $O(n)$ arithmetic operations by starting with player 1 and adding members in order of productivity until (30) fails:

Algorithm 1.

Initialization. Set $m = 1$ and $L_1 = \{1\}$. *Iteration.* While $m < \bar{m}$ and condition (30) holds, replace L_m with $L_{m+1} = L_m \cup \{m+1\}$ and increment m by 1. *Termination.* Stop and return L_m.

Part (iii) of Proposition 5 shows that a stronger taste for conformity monotonically shrinks the optimal leader group. Although one might expect stronger peer pressure to call for more leaders to shift the social norm, a higher λ makes followers more responsive to the leaders' efforts while tightening mutual norm constraints among co-leaders. Retaining more followers therefore becomes more valuable, and the planner reassigns the least productive leaders to the follower group.

This feedback erosion is sharper under social conformity than in the local-aggregate benchmark of Zhou and Chen (2015), where homogeneous complete networks obey a "50–50 rule" ($m^* = \lfloor n/2 \rfloor$). In K_n, the local-aggregate game (22) admits a finite equilibrium for all leader groups when $\lambda < \bar{\lambda} := \left(\sqrt{n^2+4n-4} - n + 2\right)/4$.[15] Within this common parameter range, both

[15] Across leader groups in K_n, the binding admissibility condition is the singleton requirement $1 - 2\lambda c_i > 0$. Since a single leader has $c_i = 1/(\eta - n + 1)$ with $\eta = (n - 1 + \lambda)/\lambda$, the condition $\eta - n + 1 > 2\lambda$ is equivalent to $\lambda < \bar{\lambda}$.

models rank candidates by productivity, but social conformity leads the planner to appoint weakly fewer leaders.

Corollary 3 (Comparison with local-aggregate interactions)**.** *In the complete network K_n with $\alpha_1 > \cdots > \alpha_n > 0$ and $0 < \lambda < \bar{\lambda}$, an optimal leader group in the local-average model is a subset of an optimal leader group in the local-aggregate model.*

Proof: See Appendix A.14.

3.4.2 Complete bipartite networks

Many organizations divide naturally into two interacting groups, such as managers and workers or upstream and downstream divisions, where links run exclusively across sides. In the complete bipartite network K_{n_1,n_2}, the player set N is partitioned into parts N_1 and N_2 of sizes $n_1 := |N_1|$ and $n_2 := |N_2|$, with $g_{ij} = 1$ if and only if players i and j belong to different parts; the star network corresponds to the special case $\min\{n_1, n_2\} = 1$. Each player's social norm is then the average effort of the opposite part. For $p \in \{1, 2\}$, let $\bar{\alpha}_p := n_p^{-1} \sum_{i \in N_p} \alpha_i$ and $\hat{\alpha}_p := \max_{i \in N_p} \alpha_i$ denote the average and maximum productivities in part N_p.

Proposition 6 (Leader selection in complete bipartite networks)**.** *In the complete bipartite network K_{n_1,n_2} with parts N_1 and N_2, suppose that productivities are heterogeneous.*

(i) Key leader. *A key leader is the most productive player in either N_1 or N_2. If $n_1 = n_2$, a key leader is the most productive player in a part N_p that maximizes $(1+\lambda)\hat{\alpha}_p + (1-\lambda)\bar{\alpha}_p$. If $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\bar{\alpha}_1 = \bar{\alpha}_2$, a key leader is the most productive player in a part of smaller size.*

(ii) Key group. *Suppose that players in each part N_p have productivity α_p. An optimal leader group consists only of players in the more productive part N_p. Define $\lambda_p := \sqrt{n_p/(n_p+1)}$. If $0 < \lambda < \lambda_p$, the unique optimal leader group is the entire part N_p. If $\lambda_p < \lambda < 1$, the optimal leader groups are the singleton subsets of N_p. Otherwise, any nonempty subset of N_p is optimal.*

Proof: See Appendix A.15.

Within each part, players occupy symmetric positions and face the same social norm, so the leading index increases with individual productivity and the key leader must be the most productive member of N_1 or N_2. When the two parts have the same size ($n_1 = n_2$), part (i) shows that selection between the two sides depends on both the maximum productivity $\hat{\alpha}_p$ and the average productivity $\bar{\alpha}_p$. Because each player's social norm is the average effort of the opposite part, a lower average productivity in N_2 than in N_1 depresses equilibrium effort in N_2 and lowers the social norm faced by members of N_1. That wider gap between productivity and the opposite part's social norm (and hence Nash effort) can make the most productive member of N_1 a more effective leader than a more productive individual in N_2 ($\hat{\alpha}_2 > \hat{\alpha}_1$), although a stronger taste for conformity shifts weight in $(1+\lambda)\hat{\alpha}_p + (1-\lambda)\bar{\alpha}_p$ from the part average $\bar{\alpha}_p$ toward the individual maximum $\hat{\alpha}_p$. When the two parts instead share the same average and maximum productivities, part (i) implies that the planner should "start small" (Zhou and Chen (2015)): each follower in the larger part has fewer neighbors and places greater weight on a leader from the smaller part, which generates stronger norm feedback. In particular, in a star network ($\min\{n_1, n_2\} = 1$), the central hub moves first.

Part (ii) turns to group selection when each part N_p has a common productivity α_p. Only members of the more productive part N_p then have productivity above the social norm formed by the opposite part ($\alpha_p > \bar{x}_i^N$, and hence $\alpha_p > x_i^N$), so Proposition 3(i) rules out leaders from the less productive part. In the local-aggregate model of Zhou and Chen (2015), the absence of links within N_p implies that the planner should use the natural bipartite partition and appoint the entire part N_p. Under social conformity, however, leaders on the same side still compete indirectly for follower feedback: moving additional members of N_p to Stage 1 leaves fewer followers in N_p to reinforce the Stage 2 responses of the opposite part, which weakens the norm feedback to leaders in N_p. When conformity is weak ($\lambda < \lambda_p$), the direct effect of having more high-productivity players commit early prevails, and the entire part N_p leads. When conformity is strong ($\lambda > \lambda_p$), the loss of indirect feedback outweighs the gain from additional commitments, and the optimal leader group reduces to a single leader in N_p.

4 Conclusion

This paper studies the optimal design of leading by example when players care about conforming to the average effort of their network neighbors. Conformity plays a dual role in organizations:

it makes followers receptive to an early example, yet it also restrains leaders from deviating from their peers. Moving first links the two roles. Because followers adjust to the leading effort, a leader shifts her own social norm and thereby alters the pressure to conform that governs the leader's choice.

This norm feedback effect determines who should lead alone. A single leader raises aggregate effort if and only if individual productivity exceeds the leader's Nash effort, and the local-average leading index can select a different key leader from the local-aggregate benchmark even when the two models share the same simultaneous-move best responses. As the taste for conformity strengthens, the selection criterion shifts from local productivity advantages weighted by two-step feedback to the ability of the most productive player to pull every effort in the network toward the maximum productivity.

The same effect governs which members should lead together. Each additional leader supplies another early example but leaves fewer followers to return feedback to the existing leaders. An optimal leader group therefore includes only players whose productivity exceeds their Nash effort, and the remaining followers form a dominating set. Under weak conformity, group selection becomes a weighted maximum directed cut problem that penalizes appointing adjacent candidates together; under strong conformity, the optimal group shrinks to a single player with the highest productivity. On complete networks, the optimal leader group contracts as conformity strengthens and never exceeds half the team. On complete bipartite networks with the same average and maximum productivities on both sides, the key leader belongs to the smaller side, whose members receive more concentrated attention from their followers.

More broadly, effective leadership design must coordinate who acts first with who remains available to respond. When leaders are themselves bound by social norms, a productive member can contribute more to collective performance by waiting and responding to colleagues than by moving first.

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A Proofs

A.1 Proof of Lemma 1

Proof. Since $\widehat{\mathbf{G}}$ is row-stochastic and $0 \leq \lambda < 1$, the inverse $(\mathbf{I} - \lambda\widehat{\mathbf{G}})^{-1} = \sum_{t=0}^{\infty} \lambda^t \widehat{\mathbf{G}}^t$ exists and is nonnegative. The best-response system has the unique solution (6). This solution is strictly positive because $\boldsymbol{\alpha} > \mathbf{0}$ and the series includes the identity matrix. Each row of $(1-\lambda)(\mathbf{I}-\lambda\widehat{\mathbf{G}})^{-1}$ sums to one. For $0 < \lambda < 1$, connectedness provides a positive-weight path between any two players, so the influence matrix has strictly positive entries. □

A.2 Proof of Theorem 1

We first establish the follower response and the bounds on norm feedback used in the leaders' equilibrium system.

Lemma 2 (Follower responses and feedback)**.** *For a leader group L with $0 < |L| < n$, $0 \leq \lambda < 1$, and nonnegative $\mathbf{x}_L$, the follower subgame admits a unique strictly positive equilibrium, given by (9). The final leader norms are given by (12). Both $\lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL}$ and $\mathbf{C}_L$ are nonnegative, $0 \leq c_i \leq \lambda$, and $\mathbf{C}_L \mathbf{1}_L \leq \mathbf{1}_L$.*

Proof. For given leading efforts $\mathbf{x}_L$, the follower best responses form the linear system

$$(\mathbf{I}_F - \lambda\widehat{\mathbf{G}}_{FF})\mathbf{x}_F = (1-\lambda)\boldsymbol{\alpha}_F + \lambda\widehat{\mathbf{G}}_{FL}\mathbf{x}_L.$$

Since $\widehat{\mathbf{G}}_{FF}$ is substochastic and $\lambda < 1$, the inverse exists and equals the nonnegative series $\mathbf{U}_F = \sum_{t\geq 0} \lambda^t \widehat{\mathbf{G}}_{FF}^t$. The identity term and positive productivities make the unique solution strictly positive for any nonnegative $\mathbf{x}_L$. Strict concavity of each follower's payoff then gives the unique follower equilibrium in (9). Its derivative with respect to $\mathbf{x}_L$ gives the follower response matrix. Substitution into $\bar{\mathbf{x}}_L = \widehat{\mathbf{G}}_{LL}\mathbf{x}_L + \widehat{\mathbf{G}}_{LF}\mathbf{x}_F$ gives the leader norms in (12) and the norm feedback matrix in (13).

To bound norm feedback, increase one leading effort by one unit and hold the other leading efforts fixed. Each follower derivative is nonnegative. Each response equation expresses that derivative as λ times a weighted average of follower derivatives and leader derivatives in $\{0, 1\}$. If the maximum follower derivative exceeded one, the corresponding response equation would contradict $\lambda < 1$. All follower derivatives are therefore at most one. Substitution into the

response equations sharpens the bound to λ. Because the network has no self-loops, c_i is a weighted sum of these follower derivatives, with nonnegative weights that sum to at most one. Hence $0 \leq c_i \leq \lambda$.

Finally, an increase of one unit in all leading efforts yields follower derivatives at most λ. The resulting change in each leader's social norm is therefore at most one, so $\mathbf{C}_L \mathbf{1}_L \leq \mathbf{1}_L$. □

With Lemma 2 established, we solve the leaders' equilibrium system and characterize the influence weights.

Proof. Step 1: Equilibrium efforts. For a leader group L, substitute the follower response (9) into the leaders' payoffs. Leader i's deviation from the social norm changes at rate $1 - c_i > 0$ with respect to own effort. The reduced payoff is strictly concave, with second derivative $-1 - \gamma(1 - c_i)^2 < 0$. At zero effort, the marginal payoff is positive: the direct term is $\alpha_i > 0$, and the norm generated by other leaders and follower productivities is nonnegative. Each best response is therefore interior, so an equilibrium must satisfy (11).

By (12), the leaders' first-order conditions become

$$\begin{aligned}&\left[(1-\lambda)\mathbf{I}_L + \lambda(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})(\mathbf{I}_L - \mathbf{C}_L)\right]\mathbf{x}_L^S \\ &\qquad = (1-\lambda)\left[\boldsymbol{\alpha}_L + \lambda(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\boldsymbol{\alpha}_F\right].\end{aligned}$$

The coefficient matrix has nonpositive off-diagonal entries. In row i, the diagonal exceeds the sum of the absolute off-diagonal entries by

$$(1-\lambda) + \lambda(1-c_i)\left(1 - \sum_{j \in L} c_{ij}\right) > 0.$$

To obtain the inverse, divide each row by its positive diagonal entry. The original coefficient matrix therefore factors as a positive diagonal matrix times $\mathbf{I} - \mathbf{B}$, where $\mathbf{B} \geq 0$ has row sums below one. The convergent series for $(\mathbf{I} - \mathbf{B})^{-1}$ gives a nonnegative inverse. Since the right-hand side is strictly positive, the system has a unique positive solution. The follower response evaluated at this solution gives (17).

Step 2: Influence weights. Let $\mathbf{E}^L$ be diagonal, with entry $1 - c_i$ for a leader and entry one for a follower. We can write the full equilibrium system as

$$\mathbf{Q}_L \mathbf{x}^S(L) = \boldsymbol{\alpha}, \qquad \mathbf{Q}_L := \mathbf{I} + \gamma \mathbf{E}^L(\mathbf{I} - \widehat{\mathbf{G}}). \tag{31}$$

The row-dominance argument in Step 1 also gives a nonnegative inverse for $\mathbf{Q}_L$. For strict positivity, suppose that $0 < \lambda < 1$. After division of row i by $1 + \gamma(\mathbf{E}^L)_{ii}$, the coefficient matrix becomes $\mathbf{I} - \mathbf{T}\widehat{\mathbf{G}}$, with

$$\mathbf{T}_{ii} := \frac{\gamma(\mathbf{E}^L)_{ii}}{1 + \gamma(\mathbf{E}^L)_{ii}} \in (0, 1).$$

Connectedness makes the series $\sum_{t \geq 0}(\mathbf{T}\widehat{\mathbf{G}})^t$ strictly positive, and hence $\mathbf{M}^S = \mathbf{Q}_L^{-1}$ is strictly positive. To obtain the row sums, use $(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{1} = \mathbf{0}$ in (31). We obtain $\mathbf{Q}_L\mathbf{1} = \mathbf{1}$, so $\mathbf{M}^S\mathbf{1} = \mathbf{1}$. At $\lambda = 0$, $\mathbf{M}^S = \mathbf{I}$. Strict concavity and interior best responses establish that the resulting outcome is the unique subgame-perfect equilibrium. □

A.3 Proof of Proposition 1

Proof. To compare the two effort profiles, use $\mathbf{E}^L$ and $\mathbf{Q}_L$ from (31), with $\mathbf{M}^S = \mathbf{Q}_L^{-1}$. The simultaneous-move first-order conditions give $\gamma(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{x}^N = \boldsymbol{\alpha} - \mathbf{x}^N$. The difference between the equilibrium systems is therefore

$$\begin{aligned}\mathbf{Q}_L(\mathbf{x}^S - \mathbf{x}^N) &= \boldsymbol{\alpha} - \mathbf{x}^N - \gamma\mathbf{E}^L(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{x}^N \\ &= (\mathbf{I} - \mathbf{E}^L)(\boldsymbol{\alpha} - \mathbf{x}^N).\end{aligned}$$

After multiplication by $\mathbf{M}^S$, coordinate j is

$$x_j^S - x_j^N = \sum_{i \in L} m_{ji}^S c_i(\alpha_i - x_i^N),$$

because the diagonal entries of $\mathbf{I} - \mathbf{E}^L$ are c_i for leaders and zero for followers. The identity $x_i^N = \sum_{j=1}^n m_{ij}^N \alpha_j$ expresses the gap in terms of productivities. Comparison with zero gives (18). □

A.4 Proof of Remark 1

Proof. Given a single leader $L = \{i\}$, the identity in the proof of Proposition 1 becomes $x_j^S - x_j^N = m_{ji}^S c_i(\alpha_i - x_i^N)$ for each player $j \in N$. Theorem 1 gives $m_{ji}^S > 0$. Connectedness

ensures that leader i has a neighbor, and all of the leader's neighbors are followers. Since the network is undirected, (14) implies

$$c_i \geq \lambda \sum_{j \in F} \widehat{g}_{ij} \widehat{g}_{ji} > 0,$$

if $\lambda > 0$. Thus the effort changes all have the sign of $\alpha_i - x_i^N$. Equation (8) gives the equivalent comparison with $x_i^N - \bar{x}_i^N$.

To obtain (19), we use the Nash equilibrium representation in (6). Note that $x_i^N = \sum_{j=1}^n m_{ij}^N \alpha_j$, we get

$$\alpha_i - x_i^N = (1 - m_{ii}^N) \left(\alpha_i - \sum_{j \neq i} \frac{m_{ij}^N}{1 - m_{ii}^N} \alpha_j \right).$$

Since $\mathbf{M}^N$ is strictly positive and row-stochastic, we have $0 < m_{ii}^N < 1$. The inequality $\alpha_i - x_i^N > 0$ is therefore equivalent to (19). □

A.5 Proof of Proposition 2

Proof. Step 1: Follower responses and norm feedback. Let player i be the sole leader. The matrix $\widehat{\mathbf{G}}_{-i}$ deletes row and column i from $\widehat{\mathbf{G}}$, and $\widehat{\mathbf{g}}_{i\cdot}$ and $\widehat{\mathbf{g}}_{\cdot i}$ denote the retained row and column, with their original normalization. By Lemma 2, the follower response satisfies

$$\frac{\partial \mathbf{x}_{-i}}{\partial x_i} = \lambda (\mathbf{I}_{-i} - \lambda \widehat{\mathbf{G}}_{-i})^{-1} \widehat{\mathbf{g}}_{\cdot i}.$$

The follower rows of column i in $(\mathbf{I} - \lambda \widehat{\mathbf{G}}) \mathbf{M}^N = (1 - \lambda) \mathbf{I}$ imply

$$(\mathbf{I}_{-i} - \lambda \widehat{\mathbf{G}}_{-i}) \mathbf{M}_{-i,i}^N = \lambda \widehat{\mathbf{g}}_{\cdot i} m_{ii}^N.$$

The follower response rates are therefore the follower entries of column i of $\mathbf{M}^N$, divided by m_{ii}^N. The leader row of the same matrix identity gives $m_{ii}^N - \lambda \widehat{\mathbf{g}}_{i\cdot} \mathbf{M}_{-i,i}^N = 1 - \lambda$, which yields the feedback formula in (20). Connectedness and undirectedness give a positive return through a follower neighbor, so $c_i > 0$. Lemma 2 bounds the feedback coefficient below one.

Step 2: The leading index. We evaluate the leader's incentive to adjust effort at the Nash benchmark. At x_i^N, the derivative of the reduced payoff is

$$\alpha_i - x_i^N - \gamma (1 - c_i)(x_i^N - \bar{x}_i^N) = c_i (\alpha_i - x_i^N),$$

where the equality follows from (8). The reduced payoff has constant second derivative $-1-\gamma(1-c_i)^2$. The leader's effort adjustment and the follower response from Step 1 therefore give

$$\mathbf{x}^S(\{i\}) - \mathbf{x}^N = \frac{c_i(\alpha_i - x_i^N)}{1+\gamma(1-c_i)^2}\frac{\mathbf{M}_{\cdot i}^N}{m_{ii}^N}. \tag{32}$$

The sum of the coordinates in (32), together with $\gamma = \lambda/(1-\lambda)$, gives (21). With $c_i = (m_{ii}^N + \lambda - 1)/(\lambda m_{ii}^N)$, the index can also be written as

$$\ell_i = \frac{(m_{ii}^N + \lambda - 1)\sum_{j=1}^n m_{ji}^N}{\lambda(m_{ii}^N)^2 + (1-\lambda)(1-m_{ii}^N)^2}(\alpha_i - x_i^N).$$

Since the benchmark X^N is common to all candidates, a key leader maximizes this index.

Step 3: The final social norm. We also express the index in terms of the leader's final social norm:

$$\ell_i = \gamma c_i \left(\sum_{j=1}^n m_{ji}^N\right)(x_i^S - \bar{x}_i^S) = \left(\frac{1}{1-\lambda} - \frac{1}{m_{ii}^N}\right)\left(\sum_{j=1}^n m_{ji}^N\right)(x_i^S - \bar{x}_i^S), \tag{33}$$

where all sequential quantities refer to $L = \{i\}$. The follower response implies

$$x_i^S - \bar{x}_i^S = x_i^N - \bar{x}_i^N + (1-c_i)(x_i^S - x_i^N).$$

By (32) and $\alpha_i - x_i^N = \gamma(x_i^N - \bar{x}_i^N)$, we obtain

$$x_i^S - \bar{x}_i^S = \frac{\lambda m_{ii}^N}{\lambda(m_{ii}^N)^2 + (1-\lambda)(1-m_{ii}^N)^2}(x_i^N - \bar{x}_i^N), \qquad x_i^S - x_i^N = \frac{m_{ii}^N + \lambda - 1}{1-\lambda}(x_i^S - \bar{x}_i^S).$$

The aggregate effort change in (32) is

$$\ell_i = \frac{\sum_{j=1}^n m_{ji}^N}{m_{ii}^N}(x_i^S - x_i^N),$$

which proves (33). □

A.6 Proof of Corollary 1

Proof. Part (i): Weak conformity. As $\lambda \downarrow 0$, the absence of self-loops gives

$$m_{ii}^N + \lambda - 1 = \lambda^2 r_i + \mathcal{O}(\lambda^3), \qquad \sum_{j=1}^n m_{ji}^N = 1 + \mathcal{O}(\lambda).$$

The productivity gap satisfies $\alpha_i - x_i^N = \lambda(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^2)$. Substituting these expansions into (21) yields

$$\ell_i(\lambda) = \lambda^2 r_i(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^3), \tag{34}$$

so a key leader maximizes $r_i(\alpha_i - \bar{\alpha}_i)$ for sufficiently small $\lambda > 0$.

Part (ii): Strong conformity. The random-walk matrix of a connected undirected graph is similar to a symmetric matrix and has a simple eigenvalue one. The stationary vector is $\boldsymbol{\pi}$, with $\pi_i = d_i / \sum_{j=1}^n d_j$. The remaining eigenvalues differ from one, including a possible eigenvalue minus one. By spectral decomposition,

$$\mathbf{M}^N = \mathbf{1}\boldsymbol{\pi}^\top + \mathcal{O}(1-\lambda). \tag{35}$$

Thus $m_{ii}^N = \pi_i + \mathcal{O}(1-\lambda)$ and $\sum_{j=1}^n m_{ji}^N = n\pi_i + \mathcal{O}(1-\lambda)$. By (6), the productivity gap is $\alpha_i - x_i^N = \alpha_i - \bar{\alpha}_g + \mathcal{O}(1-\lambda)$. The coefficient on this gap in (21) is $n + \mathcal{O}(1-\lambda)$, which proves the limiting gain in part (ii).

The same expansions also give the individual consensus limit used in the group comparison. Equation (20) implies $c_i = 1 + \mathcal{O}(1-\lambda)$, while $m_{ji}^N / m_{ii}^N = 1 + \mathcal{O}(1-\lambda)$ and $\gamma(1-c_i)^2 = \mathcal{O}(1-\lambda)$. Equations (32) and (35) then yield

$$\mathbf{x}^S(\{i\}) = \alpha_i \mathbf{1} + \mathcal{O}(1-\lambda). \tag{36}$$

There are finitely many candidates, so a strict leading-order ranking near either endpoint remains strict on a common neighborhood. This gives the stated selection implications, with the most productive players tied in the strong-conformity limit. □

A.7 Proof of Corollary 2

Proof. Part (i): Productivities. In (21), ℓ_i is a positive network coefficient, independent of productivities, multiplied by $\alpha_i - x_i^N$. Since $x_i^N = \sum_{j=1}^n m_{ij}^N \alpha_j$ with $0 < m_{ii}^N < 1$ and $m_{ij}^N > 0$ for $j \neq i$, $\partial \ell_i / \partial \alpha_i$ is proportional to $1 - m_{ii}^N > 0$ and $\partial \ell_i / \partial \alpha_j$ $(j \neq i)$ is proportional to $-m_{ij}^N < 0$.

Part (ii): Conformity. By Corollary 1, ℓ_i extends continuously to $[0,1]$. If $\bar{\alpha}_i < \alpha_i \leq \bar{\alpha}_g$, then $\ell_i(0) = 0$, ℓ_i is strictly positive for sufficiently small positive λ, and $\ell_i(1) \leq 0$. Compactness gives a global maximum, and these signs place the maximum in $(0,1)$. The index cannot

be nonincreasing from zero or nondecreasing toward a nonpositive endpoint, so the index is nonmonotone. Under the reversed inequalities, the same argument with reversed signs gives an interior global minimum and proves the second conclusion.

Part (iii): Follower links. Consider a new link $\{j,k\}$ between a neighbor $j \in N(i)$ and another follower $k \neq i,j$. The link leaves $N(i)$, d_i, and $\alpha_i - \bar{\alpha}_i > 0$ unchanged. The two endpoint degrees rise by one. The change in d_k enters r_i only if $k \in N(i)$. Hence local reciprocity $r_i = d_i^{-1}\sum_{u\in N(i)} d_u^{-1}$ decreases by $d_i^{-1}[d_j^{-1}(d_j+1)^{-1} + \mathbf{1}_{\{k\in N(i)\}} d_k^{-1}(d_k+1)^{-1}] > 0$. By (34), $\ell_i(\lambda)$ strictly decreases for all sufficiently small $\lambda > 0$. □

A.8 Proof of Proposition 3

The following lemma gives the effort comparison used to establish leader eligibility in Proposition 3. Throughout this subsection, $0 < \lambda < 1$. For the empty group, set $\mathbf{E}^{\varnothing} = \mathbf{I}$ and $\mathbf{x}^S(\varnothing) = \mathbf{x}^N$.

Lemma 3 (Leader removal)**.** *For a leader group L with $0 < |L| < n$, let*

$$B := \left\{ i \in L : \alpha_i \leq \sum_{j=1}^{n} m_{ij}^S(L)\alpha_j \right\}, \qquad A := L \setminus B.$$

If A and B are both nonempty, then $\mathbf{x}^S(A) > \mathbf{x}^S(L)$ coordinatewise. If $A = \varnothing$, then $\mathbf{x}^S(L) \leq \mathbf{x}^N$ coordinatewise.

Proof. We evaluate the equilibrium system for A at the effort profile under L. By the proof of Theorem 1, the matrix in (31) has a strictly positive inverse. For a retained leader $i \in A$, norm feedback under L has the return-walk representation

$$c_i(L) = \sum_{t\geq 0} \lambda^{t+1} \widehat{\mathbf{G}}_{iF} \widehat{\mathbf{G}}_{FF}^{t} \widehat{\mathbf{G}}_{Fi}. \tag{37}$$

When the players in B become followers, the return walks available under L remain available under A, and the additional walks have nonnegative weights. Thus $c_i(A) \geq c_i(L)$ for $i \in A$.

By (31), the difference between the two systems at $\mathbf{x}^S(L)$ is

$$(\mathbf{Q}_A \mathbf{x}^S(L) - \boldsymbol{\alpha})_i = \gamma\Big[(\mathbf{E}^A)_{ii} - (\mathbf{E}^L)_{ii}\Big](x_i^S(L) - \bar{x}_i^S(L)).$$

For $i \in B$, the coefficient change is $c_i(L) \geq 0$, and the deviation from the social norm under L is nonpositive by (11). For $i \in A$, the coefficient change is nonpositive and the deviation under L is strictly positive. The coefficients of the original followers remain unchanged. Hence $\mathbf{Q}_A \mathbf{x}^S(L) \leq \boldsymbol{\alpha}$. Since the inverse is positive, we obtain $\mathbf{x}^S(A) \geq \mathbf{x}^S(L)$.

It remains to establish strict improvement when both sets are nonempty. Take a shortest path between A and B; all internal vertices lie in F. From its endpoint $i \in A$, follow the path to B and return along the same path. Since the network is undirected and $0 < \lambda < 1$, this return walk has positive weight and is available under A but not under L. Hence $c_i(A) > c_i(L)$. The retained leader's deviation from the social norm under L is strictly positive, so at least one coordinate of $\boldsymbol{\alpha} - \mathbf{Q}_A \mathbf{x}^S(L)$ is strictly positive. The strictly positive inverse then gives a strict increase in each player's effort. If $A = \varnothing$, the weak comparison gives $\mathbf{x}^N \geq \mathbf{x}^S(L)$. □

We now return to the proof of Proposition 3.

Proof. We first show that an optimal leader group is nonempty and proper. For a player i with the highest productivity, positivity of the simultaneous-move influence weights and heterogeneous productivities imply

$$\alpha_i - x_i^N = \sum_{j=1}^{n} m_{ij}^N (\alpha_i - \alpha_j) > 0.$$

By Proposition 2, $\ell_i > 0$. The empty and full groups yield zero gain, so neither can be optimal.

Part (i): Leader eligibility. Lemma 3 excludes an optimal group in which all leaders exert effort at or above productivity, as well as a group combining such leaders with leaders whose efforts are below productivity. Thus $\alpha_i - x_i^S(L^*) > 0$ for each leader $i \in L^*$. To compare equilibrium efforts with the Nash benchmark, we subtract the simultaneous-move system from (31). The leader conditions then give

$$\mathbf{x}^S(L^*) - \mathbf{x}^N = \sum_{i \in L^*} \frac{c_i(L^*)}{1 - c_i(L^*)} \mathbf{M}_{\cdot i}^N [\alpha_i - x_i^S(L^*)]. \tag{38}$$

The terms in (38) are nonnegative. Connectedness ensures that at least one leader has a follower neighbor, and the two-step return through that neighbor gives strictly positive norm feedback. Since $\mathbf{M}^N$ has positive entries, $\mathbf{x}^S(L^*) > \mathbf{x}^N$. Together with the upper bound $x_i^S(L^*) < \alpha_i$ and (8), this yields $\bar{x}_i^N < x_i^N < x_i^S(L^*) < \alpha_i$ for each leader $i \in L^*$, so Proposition 2 implies $\ell_i > 0$. This proves part (i).

Part (ii): Follower neighbors. Suppose, to the contrary, that a leader $i \in L^*$ has no follower neighbor. Then $c_i(L^*) = 0$. By connectedness, player i has a neighbor $j \in L^*$, so $R := L^* \setminus \{i\}$ is nonempty. Under R, player i becomes a follower. The corresponding coefficient in $\mathbf{E}$ remains one, and the coefficients of the original followers do not change. The return walks available to a retained leader under L^* remain available under R, so $c_k(R) \geq c_k(L^*)$. The additional walk $j \to i \to j$ makes the inequality strict for j.

By the upper effort bound in part (i) and (11), $(x_k^S(L^*) - \bar{x}_k^S(L^*)) > 0$ for all retained leaders. With the changed coefficients, the original effort profile satisfies

$$\mathbf{Q}_R \mathbf{x}^S(L^*) \leq \boldsymbol{\alpha}, \qquad (\mathbf{Q}_R \mathbf{x}^S(L^*))_j < \alpha_j.$$

Since $\mathbf{Q}_R^{-1} > 0$, the nonnegative residual with a strictly positive coordinate implies $\mathbf{x}^S(R) > \mathbf{x}^S(L^*)$. This contradicts the optimality of L^*. Thus each leader in an optimal group has a follower neighbor.

Under a positive upper bound on group size, the singleton led by a player with the highest productivity and each leader deletion remain feasible. The preceding comparisons therefore establish the same restrictions and effort bounds under that constraint. □

A.9 Proof of Proposition 4

The following lemma identifies the limiting effort profile for a fixed leader group. We use this limit to compare groups in part (ii) of Proposition 4.

Lemma 4 (Consensus under leadership)**.** *For a leader group L with $2 \leq |L| < n$, each $c_i^\infty < 1$, and $\mathbf{x}^S(L)$ converges to the scalar in (28) times $\mathbf{1}$. A singleton instead satisfies (36). The empty and full groups converge to $\bar{\alpha}_g \mathbf{1}$.*

Proof. For a group with at least two leaders, each follower has a positive-probability random-walk path to a leader. Finiteness gives a common finite horizon and a positive lower bound on the probability of leaving the follower set. Thus the substochastic matrix $\widehat{\mathbf{G}}_{FF}$ is transient and has spectral radius below one. Hence

$$\mathbf{C}_L^\infty := \lim_{\lambda \to 1} \mathbf{C}_L = \widehat{\mathbf{G}}_{LL} + \widehat{\mathbf{G}}_{LF}(\mathbf{I}_F - \widehat{\mathbf{G}}_{FF})^{-1}\widehat{\mathbf{G}}_{FL}$$

is well defined. Entry (i, j) is the probability that a walk starting at leader i next reaches the leader set at j. Each leader has a path to another leader with no internal leader. Since that path has positive probability, $1 - c_i^\infty > 0$.

We next show that any limit of equilibrium efforts is a consensus profile. By Theorem 1, all efforts lie between the minimum and maximum productivities. The feedback limit bounds the diagonal entries of $\mathbf{E}^L$ away from zero as λ approaches one. Equation (31) can be written as $(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{x}^S = \gamma^{-1}(\mathbf{E}^L)^{-1}(\boldsymbol{\alpha} - \mathbf{x}^S) = \mathcal{O}(\gamma^{-1})$. A convergent subsequence therefore has a limit in the null space of $\mathbf{I} - \widehat{\mathbf{G}}$, which is spanned by $\mathbf{1}$.

To determine the common effort level, multiply (31) by $\boldsymbol{\pi}^\top(\mathbf{E}^L)^{-1}$. Since $\boldsymbol{\pi}^\top(\mathbf{I} - \widehat{\mathbf{G}}) = \mathbf{0}^\top$, we obtain

$$\sum_{i \in L} \frac{\pi_i}{1 - c_i} x_i^S + \sum_{j \in F} \pi_j x_j^S = \sum_{i \in L} \frac{\pi_i}{1 - c_i} \alpha_i + \sum_{j \in F} \pi_j \alpha_j. \tag{39}$$

Passing to the limit gives (28). All convergent subsequences have this limit times $\mathbf{1}$, so the full profile converges. For a singleton, $c_i^\infty(\{i\}) = 1$ and (36) gives the limit. The empty and full groups reproduce the Nash equilibrium, so (35) applies. □

With Lemma 4 established, we prove the weak- and strong-conformity statements in turn.

Proof. Part (i): Weak conformity. The expansion gives a common objective for group selection. We then use that objective to screen candidates and characterize when appointing all candidates is optimal.

Step 1: The aggregate gain. For a nonempty proper group L, define

$$\mathbf{H}_L := \mathbf{I}_L + \gamma(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})(\mathbf{I}_L - \mathbf{C}_L).$$

The leader system and the Nash equilibrium can be evaluated along the same follower response. Their difference is

$$\mathbf{x}_L^S - \mathbf{x}_L^N = \mathbf{H}_L^{-1}\mathbf{C}_L^{\mathrm{D}}(\boldsymbol{\alpha}_L - \mathbf{x}_L^N).$$

The follower effort difference equals $\lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL}$ times the leader effort difference. To obtain the leading term in the aggregate gain, we use the following expansions near zero:

$$\mathbf{H}_L^{-1} = \mathbf{I}_L + \mathcal{O}(\lambda),$$

$$\begin{aligned}
\mathbf{C}_L^{\mathrm{D}} &= \lambda(\widehat{\mathbf{G}}_{LF}\widehat{\mathbf{G}}_{FL})^{\mathrm{D}} + \mathcal{O}(\lambda^2),\\
\boldsymbol{\alpha}_L - \mathbf{x}_L^N &= \lambda[(\mathbf{I} - \widehat{\mathbf{G}})\boldsymbol{\alpha}]_L + \mathcal{O}(\lambda^2),\\
\mathbf{1}_L^\top + \lambda \mathbf{1}_F^\top \mathbf{U}_F \widehat{\mathbf{G}}_{FL} &= \mathbf{1}_L^\top + \mathcal{O}(\lambda).
\end{aligned}$$

The product of these expressions gives $\ell(L) = \lambda^2\Lambda(L) + \mathcal{O}(\lambda^3)$, where $\Lambda(L)$ is defined in (25). The empty and full groups have zero gain and zero coefficient. Since the number of groups is finite, the remainder is uniform across groups, and a strict difference in the leading coefficients remains strict for sufficiently small positive λ.

Step 2: Candidate screening. Let $A := L \cap N_+$ and $B := L \setminus N_+$. Removing B eliminates its nonpositive contributions to followers and makes its members available to respond to leaders in A. The change in the objective is

$$\Lambda(A) - \Lambda(L) = \sum_{i\in A, j\in B} r_{ij}(\alpha_i - \bar{\alpha}_i) - \sum_{i\in B, j\notin L} r_{ij}(\alpha_i - \bar{\alpha}_i) \geq 0.$$

Hence restricting a group to N_+ weakly increases the objective. If all local productivity gaps are nonzero and B is nonempty, each gap in B is negative. Heterogeneous productivities and $\boldsymbol{\pi}^\top(\mathbf{I} - \widehat{\mathbf{G}})\boldsymbol{\alpha} = 0$ guarantee a positive-gap player outside B. Connectedness then gives an edge from B to a player either in A or outside L. That edge makes a strictly positive contribution to the displayed difference. Thus a leading-order optimum is contained in N_+ when no local productivity gap is zero.

Step 3: Interactions among candidates. For $L \subseteq N_+$, the sum of the singleton values $r_i(\alpha_i - \bar{\alpha}_i)$ counts both directions of each link within L. Subtracting those contributions gives the pairwise-loss formula (26). For $S, T \subseteq N_+$, the singleton contributions cancel in the submodularity difference, and we obtain

$$\Lambda(S) + \Lambda(T) - \Lambda(S \cup T) - \Lambda(S \cap T) = \sum_{\substack{i\in S\setminus T\\ j\in T\setminus S}} r_{ij}[(\alpha_i - \bar{\alpha}_i) + (\alpha_j - \bar{\alpha}_j)] \geq 0.$$

Thus Λ is submodular on 2^{N_+}. The inequality is strict precisely when an edge joins $S \setminus T$ to $T \setminus S$. In particular, for $L \subseteq N_+$ and distinct $i, j \in N_+ \setminus L$, appointing j reduces the marginal contribution of appointing i by $r_{ij}[(\alpha_i - \bar{\alpha}_i) + (\alpha_j - \bar{\alpha}_j)]$, which is strictly positive when the

candidates are adjacent. Comparing each candidate's appointment with leaving that candidate as a follower gives

$$\Lambda(N_+) - \Lambda(N_+ \setminus \{i\}) = (\alpha_i - \bar{\alpha}_i) \sum_{j \notin N_+} r_{ij} - \sum_{j \in N_+} r_{ij}(\alpha_j - \bar{\alpha}_j).$$

If N_+ is optimal, each deletion margin must be nonnegative. Conversely, suppose all deletion margins are nonnegative. By submodularity, the marginal contribution of a candidate to a smaller group is at least as large as the corresponding margin at N_+. Adding the missing candidates therefore cannot lower the objective. Step 2 extends this comparison from candidate groups to all groups.

If all deletion margins are strictly positive, N_+ is uniquely optimal among candidate groups. A group omitting a candidate has a strictly smaller value after screening and hence also before screening. By the uniform expansion in Step 1, an exact optimum near zero contains N_+. If no local productivity gap is zero, the strict screening comparison also excludes additional noncandidates, so the exact optimum is unique. If a deletion margin is negative, removing that candidate strictly improves on N_+, whereas screening bounds the value of any group containing N_+ by $\Lambda(N_+)$. The strict difference persists near zero, so no exact optimum contains all candidates. Zero deletion margins allow leading-order ties.

Part (ii): Strong conformity. A singleton led by a player i with the highest productivity yields limiting aggregate effort $n \max_j \alpha_j$. We compare this limit with those of singletons with lower productivity and groups of other sizes. A singleton with lower productivity has a strictly lower limit. By Lemma 4, a proper group with at least two members has a limit equal to n times an average that assigns positive weight to each player's productivity. Heterogeneity makes that limit strictly lower as well. The empty and full groups have limit $n\bar{\alpha}_g < n \max_j \alpha_j$. There are finitely many groups of these types, so the singleton led by i strictly outperforms all of them on a common neighborhood below one. An optimal group in that neighborhood must therefore consist of a single player with the highest productivity.

For the individual effort comparison, a singleton led by a player with the highest productivity makes all efforts converge to $\max_j \alpha_j$. For a group of at least two such leaders, the common limit in Lemma 4 assigns positive weight to all productivities, including a lower-productivity follower, and is therefore strictly smaller. The number of singletons, groups and players is finite, so all coordinatewise comparisons hold on a common neighborhood below one. □

A.10 Proof of Claim 1

Proof. The submodularity claim follows from the pairwise-loss calculation in the proof of Proposition 4. We reduce MAX-CUT on connected regular graphs to (25). This source problem is NP-hard: Jelínek et al. (2026) establish NP-completeness of Large-Deg-Max-Cut, whose input graphs on n vertices are $(n-4)$-regular with connected cubic complements. For $n \geq 8$, degree $n-4 \geq n/2$ also makes the input graph connected; excluding finitely many smaller instances does not affect hardness.

Let H be such a connected d-regular graph, with adjacency matrix $\mathbf{A}$, and fix a vertex r. Write $\boldsymbol{\mathcal{L}} = d\mathbf{I} - \mathbf{A}$ for its Laplacian and $\boldsymbol{\mathcal{L}}_{-r}$ for the principal submatrix obtained by deleting r. Keep H as the interaction network and set

$$\alpha_r = 1, \qquad \boldsymbol{\alpha}_{-r} = \mathbf{1}_{-r} + d\boldsymbol{\mathcal{L}}_{-r}^{-1}\mathbf{1}_{-r}.$$

The grounded Laplacian is positive definite: extending a vector by zero at r gives the quadratic form $\sum_{\{i,j\}\in E(H)}(z_i - z_j)^2$, which vanishes only at zero by connectedness. The vector $\mathbf{u}_{-r} = d\boldsymbol{\mathcal{L}}_{-r}^{-1}\mathbf{1}_{-r}$ is strictly positive. Otherwise a nonpositive minimum, with $u_r = 0$, would satisfy $\sum_{j\in N(i)}(u_i - u_j) \leq 0$, contradicting its equation $\sum_{j\in N(i)}(u_i - u_j) = d$. Thus all productivities are positive. The construction gives

$$\alpha_i - \bar{\alpha}_i = \begin{cases} 1, & i \neq r, \\ -(n-1), & i = r. \end{cases}$$

The second equality follows because local productivity gaps sum to zero on a regular undirected graph. The rational linear system can be solved in polynomial time, and determinant bounds give polynomial bit length for the resulting productivities.

It remains to relate the leadership objective to the cut value. An optimal group leaves r as a follower: if $r \in L$, and a and b count its follower and other leader neighbors, removing r raises the objective by $[(n-1)a + b]/d^2 > 0$, since $a + b = d$. For $L \subseteq N \setminus \{r\}$, each cut edge has a leading endpoint with local productivity gap one and link reciprocity $1/d^2$. Therefore

$$\Lambda(L) = \frac{|\partial_H L|}{d^2}, \qquad \max_{L\subseteq N} \Lambda(L) = \frac{\mathrm{MAXCUT}(H)}{d^2}.$$

The second equality uses the symmetry of an undirected cut: either side of a maximum cut can be designated as leaders, so requiring r to follow loses no cut value. An instance with cut threshold k is therefore equivalent to the leadership instance with threshold k/d^2. The reduction preserves the network and has polynomial encoding length, proving the claim. □

A.11 Proof of the candidate-retention statement

Proof. If $N_+ \subseteq L^*$, the candidate i with $N(i) \subseteq N_+$ is a leader without a follower neighbor, contradicting Proposition 3(ii). □

A.12 Proof of Proposition 5

Proof. Parts (i) and (ii): Leader selection and group size. We first express the gain in terms of group size and the sum of leading productivities. For a group L of size m, let $f := n - m$, $a_L := \sum_{i \in L} \alpha_i$ and $a_N := \sum_{i=1}^{n} \alpha_i$. Each follower responds to a given leader at rate $\gamma/(n-1+m\gamma)$, so each leader's norm feedback is

$$t_m := \frac{f\gamma}{(n-1)(n-1+m\gamma)}.$$

Let $s := 1 - t_m$, $v := 1 + \gamma n/(n-1)$ and $v_L := 1 + \gamma s n/(n-1)$. The equilibrium conditions are $vx_j = \alpha_j + \gamma X/(n-1)$ for followers and $v_L x_i = \alpha_i + \gamma s X/(n-1)$ for leaders. The sum over both groups gives

$$X^S(L) = \frac{a_L v + (a_N - a_L)v_L}{1 + \gamma(m + sf)/(n-1)}.$$

Since $X^N = a_N$ on a complete network, subtraction and simplification give

$$\ell(L) = \frac{\gamma t_m}{n - 1 + \gamma(n - f t_m)}(n a_L - m a_N). \tag{40}$$

At a prescribed size, only a_L depends on group membership. Its coefficient is strictly positive, so the strict productivity ordering uniquely selects L_m and, at $m = 1$, makes player 1 the unique key leader. This establishes part (i). Substitution of t_m into (40) gives

$$\ell(L_m) = \frac{n\gamma^2(n-m)\sum_{j=1}^{m}(\alpha_j - \bar{\alpha})}{(n-1)^3 + (n-1)^2(n+m)\gamma + [n(n-1)m - (n-m)^2]\gamma^2}. \tag{41}$$

To compare group sizes, let $T_m := \sum_{j=1}^{m}(\alpha_j - \bar{\alpha})$ and $A_m := (n-m)T_m$. The strict productivity ordering implies $T_m > 0$ for $1 \le m \le n-1$. Write the denominator of (41) as

$$\begin{aligned} D_m(\gamma) &:= d_0 + b_m\gamma + a_m\gamma^2, \\ d_0 &:= (n-1)^3, \qquad b_m := (n-1)^2(n+m), \\ a_m &:= n(n-1)m - (n-m)^2. \end{aligned}$$

Then $\ell(L_m) = n\gamma^2 A_m / D_m$. Both D_m and its coefficients are positive. Since $\eta = (n-1+n\gamma)/\gamma$,

$$\frac{D_m}{\gamma^2} = (n-1)\eta(\eta - n + m) - (n-m)^2,$$

so (41) equals the objective in (29). If $\alpha_m \le \bar{\alpha}$, then $m \ge 2$, $A_m < A_{m-1}$, and $D_m > D_{m-1}$. Such a group has $\ell(L_m) < \ell(L_{m-1})$ and cannot be optimal.

We next show that the optimal size is at most $\bar{m}$. Player 1 alone yields a positive gain, whereas the empty and full groups reproduce simultaneous play, so an optimal group has size between 1 and $n-1$. Suppose a top-m group has $m > \bar{m}$ and $\alpha_m > \bar{\alpha}$. Since $T_{m-1} \ge (m-1)(\alpha_m - \bar{\alpha})$,

$$A_m - A_{m-1} = (n-m)(\alpha_m - \bar{\alpha}) - T_{m-1} \le (n-2m+1)(\alpha_m - \bar{\alpha}) \le 0.$$

Moreover, for $2 \le m \le n-1$ and $\gamma > 0$,

$$D_m - D_{m-1} = (n-1)^2\gamma + [n(n+1) - 2m + 1]\gamma^2 > 0.$$

As $A_{m-1} > 0$, these inequalities imply $\ell(L_m) < \ell(L_{m-1})$. Hence no size above $\bar{m}$ can be optimal. For $n = 2$, the only nontrivial size is one. Fixed-size productivity sorting and the bound $1 \le m^* \le \bar{m}$ therefore establish the exact reduction in part (ii).

Part (iii): Contraction with conformity. For $m < s$, it suffices to show that $\ell(L_s)/\ell(L_m)$ strictly decreases with γ. Since A_s/A_m does not depend on γ, we compare the denominators. The numerator of the derivative of D_s/D_m is

$$D_s' D_m - D_s D_m' = d_0(b_s - b_m) + 2d_0(a_s - a_m)\gamma + (a_s b_m - a_m b_s)\gamma^2.$$

All three coefficients are strictly positive, because

$$a_s - a_m = (s-m)[n(n+1) - s - m] > 0,$$

$$a_s b_m - a_m b_s = (n-1)^2(s-m)\{n^2(n+2) - n(s+m) - sm\} > 0.$$

For $m, s \leq n-1$, the last bracket is at least $n^3 - n^2 + 4n - 1$. Hence D_s/D_m strictly increases with γ, and $\ell(L_s)/\ell(L_m) = (A_s/A_m)D_m/D_s$ strictly decreases. If size m is optimal at one parameter, no larger size can be optimal at a larger parameter. Since $\gamma = \lambda/(1-\lambda)$ strictly increases with λ and optimal groups share the same strict productivity ranking, $L^*(\lambda') \subseteq L^*(\lambda)$ for $\lambda' > \lambda$. This proves part (iii). □

A.13 Proof of Claim 2

Proof. We compare adjacent group sizes and then show that the gain increments change sign at most once. For $1 \leq m \leq \bar{m}$, define

$$T_m := \sum_{i=1}^{m} (\alpha_i - \bar{\alpha}), \qquad Q_m := \frac{(n-1)\eta^2}{n-m} - (n-1)\eta - (n-m).$$

The objective in (29) is $\ell(L_m) = nT_m/Q_m$. Strict productivity ordering gives $T_m > 0$. Since $\eta > n$, $Q_1 = \eta^2 - (n-1)\eta - (n-1) > 1$, and for $1 \leq m < \bar{m}$,

$$\Delta Q_m := Q_{m+1} - Q_m = 1 + \frac{(n-1)\eta^2}{(n-m)(n-m-1)} > 0.$$

Thus all Q_m are positive, and the increments ΔQ_m strictly increase with m. Subtracting adjacent gains yields

$$\ell(L_{m+1}) - \ell(L_m) = \frac{nH_m}{Q_m Q_{m+1}}, \qquad H_m := (\alpha_{m+1} - \bar{\alpha})Q_m - T_m \Delta Q_m.$$

Since $T_m/Q_m = \ell(L_m)/n$, the inequality $H_m > 0$ is exactly (30).

It remains to compare the signs of successive gain increments. For $1 \leq m < \bar{m} - 1$, the identities $T_{m+1} = T_m + \alpha_{m+1} - \bar{\alpha}$ and $Q_{m+1} = Q_m + \Delta Q_m$ imply

$$H_{m+1} - H_m = (\alpha_{m+2} - \alpha_{m+1})Q_{m+1} - T_{m+1}(\Delta Q_{m+1} - \Delta Q_m) < 0.$$

Hence the signs of the gain increments change at most once, from positive to nonpositive, and all increments after the first nonpositive one are strictly negative. Algorithm 1 therefore stops at an optimal size. If no appointment is rejected, the gain increases through $\bar{m}$, so $L_{\bar{m}}$ is optimal. The same conclusion holds directly when $\bar{m} = 1$. Computing $\bar{\alpha}$ takes $O(n)$ operations, and updating

T_m and evaluating each appointment takes constant time. At most $\bar{m} - 1$ appointments are tested, giving $O(n)$ arithmetic operations for the stated productivity ordering. □

A.14 Proof of Corollary 3

Proof. We first identify the parameter range in which all leader groups are admissible in both models. Let $d := n - 1$, $H := d + n\gamma$ and $f := n - m$. Under the local-aggregate payoff (22), follower responses coincide with those in the local-average model, so norm feedback remains $t_m = f\gamma/[d(d + m\gamma)]$. The reduced local-aggregate leader system has nonpositive off-diagonal entries and row sum

$$\frac{dH - \gamma^2 f}{d(d + m\gamma)}.$$

The inequality $\gamma^2 < n\gamma + n - 1$ makes the row sums positive for any m, giving a unique positive equilibrium by strict diagonal dominance and positivity of the right-hand side. The bound is tight across groups: at or above the boundary, the singleton leader's reduced payoff has nonnegative curvature and a positive linear coefficient, hence no finite optimum. Thus $0 < \gamma < (n + \sqrt{n^2 + 4(n-1)})/2$ permits unrestricted group selection in both models.

Let $\mathcal{L}(L_m)$ be the gain from the top m players in the corresponding local-aggregate model. Summing the leader conditions $(1 + n\gamma/d - \gamma t_m)x_i = \alpha_i + \gamma X/d$ and the follower conditions $(1 + n\gamma/d)x_j = \alpha_j + \gamma X/d$ gives

$$\mathcal{L}(L_m) = \frac{\gamma^2(n-m)\left[d\sum_{i=1}^{m}\alpha_i + m\gamma n\bar{\alpha}\right]}{E_m}, \qquad E_m := (d + m\gamma)[dH - \gamma^2(n-m)] > 0. \tag{42}$$

For an arbitrary group of size m, the same formula replaces the top-productivity sum by the sum over that group. Its coefficient is positive, so an optimal group at a fixed size selects the highest productivities in both models. Both gains are positive for top groups of sizes $1, \ldots, n-1$, whereas the empty and full groups have zero gain.

To compare optimal group sizes, we use D_m and T_m from the proof of Proposition 5. Define $z_m := m(n-m)/D_m > 0$ and $\sigma_m := T_m/m$. Heterogeneity implies $\sigma_m > 0$, and the productivity ordering makes σ_m weakly decreasing in m. Since

$$D_m = dH(d + m\gamma) - \gamma^2(n-m)^2, \qquad \frac{E_m}{D_m} = 1 - \gamma^2 z_m\left(1 + \gamma - \frac{1}{m}\right) > 0,$$

equations (41) and (42) imply

$$\ell(L_m) = n\gamma^2 z_m \sigma_m, \qquad \mathcal{L}(L_m) = \frac{(d/n)\ell(L_m) + \gamma^2 H \bar{\alpha} z_m}{1 - \gamma^2 z_m (1 + \gamma - 1/m)}. \tag{43}$$

Suppose $s > r$ and $\ell(L_s) \geq \ell(L_r)$. Because $\sigma_s \leq \sigma_r$, the first identity gives $z_s \geq z_r$. In the second identity the numerator therefore weakly increases, while the positive denominator strictly decreases, since $1 + \gamma - 1/s > 1 + \gamma - 1/r$. Hence

$$\ell(L_s) \geq \ell(L_r) \quad \Longrightarrow \quad \mathcal{L}(L_s) > \mathcal{L}(L_r) \qquad (s > r).$$

If an optimal local-average size exceeded an optimal local-aggregate size, this implication would contradict local-aggregate optimality. Thus an optimal local-average size is no larger than any optimal local-aggregate size. Since both models choose the most productive players at a prescribed size, an optimal local-average group is contained in any optimal local-aggregate group. □

A.15 Proof of Proposition 6

Proof. Part (i): Key leader. To compare candidates, we first express the leading index in terms of individual productivity and the two part means. The mean Nash efforts in the two parts satisfy $\bar{x}_1^N = (1 - \lambda)\bar{\alpha}_1 + \lambda \bar{x}_2^N$ and $\bar{x}_2^N = (1 - \lambda)\bar{\alpha}_2 + \lambda \bar{x}_1^N$. The solution is

$$\bar{x}_1^N = \frac{\bar{\alpha}_1 + \lambda \bar{\alpha}_2}{1 + \lambda}, \qquad \bar{x}_2^N = \frac{\bar{\alpha}_2 + \lambda \bar{\alpha}_1}{1 + \lambda}.$$

For a candidate $i \in N_1$, the follower response equations imply that an increase in x_i raises each effort in N_2 at rate $\lambda/[n_1 - \lambda^2(n_1 - 1)]$ and each follower effort in N_1 at rate $\lambda^2/[n_1 - \lambda^2(n_1 - 1)]$. These responses determine the norm feedback and the network propagation multiplier:

$$c_i = \frac{\lambda}{n_1 - \lambda^2(n_1 - 1)}, \qquad \frac{\sum_{j \in N} m_{ji}^N}{m_{ii}^N} = \frac{n_1 + \lambda n_2}{n_1 - \lambda^2(n_1 - 1)}.$$

Let $H_r := r^2(1 - \lambda)(1 + \lambda)^2 + \lambda^3 > 0$ for $r \in \{n_1, n_2\}$. By (21) and $x_i^N = (1 - \lambda)\alpha_i + \lambda \bar{x}_2^N$, we obtain

$$\ell_i = \frac{\lambda^2(n_1 + \lambda n_2)}{H_{n_1}} \left(\alpha_i - \frac{\bar{\alpha}_2 + \lambda \bar{\alpha}_1}{1 + \lambda} \right), \qquad i \in N_1.$$

Interchanging n_1, n_2 and the two part means gives the expression for $j \in N_2$. Within either part, the network coefficient is strictly positive and the part means are common to all candidates. The leading index therefore ranks candidates by individual productivity. A key leader must have the highest productivity in the corresponding part, and candidates tied for the highest productivity have the same index. Let ℓ_{N_1} and ℓ_{N_2} denote the maximal indices in the two parts.

For equal part sizes, $n_1 = n_2$ implies $H_{n_1} = H_{n_2}$. The difference between the maximal indices is then

$$\ell_{N_1} - \ell_{N_2} = \frac{\lambda^2 n_1}{H_{n_1}} \left[(1+\lambda)(\hat{\alpha}_1 - \hat{\alpha}_2) + (1-\lambda)(\bar{\alpha}_1 - \bar{\alpha}_2) \right].$$

Since the coefficient is strictly positive, the optimal part maximizes $(1+\lambda)\hat{\alpha}_p + (1-\lambda)\bar{\alpha}_p$ over $p \in \{1,2\}$. Both parts are eligible at equality.

If the two parts share the same average and maximum productivities, let $\bar{\alpha}_1 = \bar{\alpha}_2 = \mu$ and $\hat{\alpha}_1 = \hat{\alpha}_2 = a$. Heterogeneous productivities imply $a > \mu$. The maximal indices are $\ell_{N_1} = \lambda^2(n_1 + \lambda n_2)(a-\mu)/H_{n_1}$ and $\ell_{N_2} = \lambda^2(n_2 + \lambda n_1)(a-\mu)/H_{n_2}$, with difference

$$\ell_{N_1} - \ell_{N_2} = \frac{\lambda^2(a-\mu)(n_2 - n_1)(1-\lambda)}{H_{n_1} H_{n_2}} \left\{ (1+\lambda)^2 \left[n_1 n_2 + \lambda(n_1^2 + n_1 n_2 + n_2^2) \right] - \lambda^3 \right\}.$$

The term in braces is strictly positive: $n_1, n_2 \geq 1$ and $0 < \lambda < 1$ imply $(1+\lambda)^2 n_1 n_2 > \lambda^3$. Hence $\ell_{N_1} - \ell_{N_2}$ has the sign of $n_2 - n_1$, and a part of smaller size is optimal. This proves part (i).

Part (ii): Key group. Suppose $\alpha_i = \alpha_p$ for $i \in N_p$ ($p \in \{1,2\}$). Heterogeneity gives $\alpha_1 \neq \alpha_2$. Let N_p be the more productive part and N_q the other part, so $\alpha_p > \alpha_q$. The Nash efforts satisfy $x_i^N = (\alpha_p + \lambda\alpha_q)/(1+\lambda) < \alpha_p$ for $i \in N_p$ and $x_j^N = (\alpha_q + \lambda\alpha_p)/(1+\lambda) > \alpha_q$ for $j \in N_q$. By Proposition 3(i), no player in N_q belongs to an optimal leader group. A singleton in N_p yields a strictly positive gain, so an optimal group is a nonempty subset $L \subseteq N_p$.

It remains to determine how many players from N_p should lead. For a nonempty group $L \subseteq N_p$ of size $k := |L| \in \{1, \ldots, n_p\}$, let $D := n_p - \lambda^2(n_p - k) > 0$. The follower response system (9) gives $c_{ij}(\lambda; L) = \lambda/D$ for all $i, j \in L$. The sum of the leader first-order conditions and the aggregate follower response give

$$\sum_{i \in L}(x_i^S(L) - x_i^N) = \frac{(\lambda/D)\sum_{i \in L}(\alpha_i - x_i^N)}{1 + \gamma(1 - \lambda/D)(1 - k\lambda/D)},$$

$$\ell(L) = \frac{n_p + \lambda n_q}{D} \sum_{i \in L}(x_i^S(L) - x_i^N).$$

For each $i \in L$, $\alpha_i - x_i^N = \lambda(\alpha_p - \alpha_q)/(1+\lambda)$. This identity and $\gamma = \lambda/(1-\lambda)$ yield

$$\ell(L) = \frac{\lambda^2(n_p + \lambda n_q)(\alpha_p - \alpha_q)}{1+\lambda} \frac{k}{a_p k + b_p}, \tag{44}$$

where $a_p := \lambda^2[n_p(1+\lambda) + \lambda] > 0$ and $b_p := n_p(1+\lambda)[n_p - (n_p+1)\lambda^2]$, with $a_p k + b_p > 0$ for all $1 \le k \le n_p$.

For $1 \le k < n_p$, the difference $(k+1)/[a_p(k+1) + b_p] - k/(a_p k + b_p) = b_p/[(a_p k + b_p)(a_p(k+1) + b_p)]$ has the sign of b_p. Since $b_p \gtrless 0$ according as $\lambda \lessgtr \sqrt{n_p/(n_p+1)} = \lambda_p$, $\ell(L)$ strictly increases with k when $0 < \lambda < \lambda_p$ and strictly decreases with k when $\lambda_p < \lambda < 1$. At $\lambda = \lambda_p$, the gain is independent of k. When $n_p = 1$, the entire part N_p and its singleton coincide. These comparisons establish the optimal leader groups in all three regimes. □